



सत्यमेव जयते
भारत सरकार
Government of India
विद्युत मंत्रालय
Ministry of Power
पूर्वी क्षेत्रीय विद्युत समिति



Eastern Regional Power Committee

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सं /NO. ERPC/EE/OPERATION/2025/2052

दिनांक/DATE: 06.03.2025

सेवा में /To
संलग्न सूची के अनुसार /As per list enclosed.

विषय : 25 फ़रवरी 2025 (मंगलवार) को कोलकाता में भौतिक रूप से आयोजित 224वीं OCC बैठक का कार्यवृत्त - संबंध में।

Sub: Minutes of 224th OCC Meeting held on 25.02.2025 (Tuesday) physically at Kolkata – reg

महोदय/महोदया,
Sir(s)/Madam,

कृपया अपनी जानकारी और आवश्यक कार्रवाई के लिए 25 फ़रवरी 2025 (मंगलवार) को कोलकाता में 10:30 बजे भौतिक रूप से आयोजित 224वीं ओसीसी बैठक के संलग्न कार्यवृत्त देखें। यह ईआरपीसी वेबसाइट (www.erpc.gov.in) पर भी उपलब्ध है।

Please find enclosed **Minutes of 224th OCC Meeting** held on **25.02.2025 (Tuesday)** physically at Kolkata at 10:30 hrs for your **kind information** and **necessary action**. The same is also available at ERPC website (www.erpc.gov.in).

टिप्पणियाँ, यदि कोई हों, कृपया यथाशीघ्र इस कार्यालय को अग्रेषित करें।

Observations, if any, may please be forwarded to this office at the earliest.

भवदीय /Yours faithfully

06.3.2025
(N.S. Mondal)

Member Secretary/सदस्य सचिव

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Chief Engineer, OPM, CEA	Chief Engineer, NPC, CEA	ASSISTANT SECRETARY,ERPC
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ERPC:: Kolkata

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ईआरपीसी:: कोलकाता



MINUTES OF 224th OCC MEETING

Date: 25.02.2025

Eastern Regional Power Committee

Contents

1. PART-A: CONFIRMATION OF MINUTES	1
1.1. Confirmation of Minutes of 223 rd OCC Meeting held on 24 th January 2025 physically at Kolkata.....	3
2. PART-B: ITEMS FOR DISCUSSION.....	3
2.1 SOP on Shifting of Transmission lines for other infrastructure projects: ERPC 3	
2.2 Grid Disturbance at 765/400kV Angul S/S, 400kV GMR and 400kV JIPL: ERPC	4
2.3 Bus split operationalization at NTPC Kahalgaon: ERPC.....	7
2.4 Reliability and Healthiness of equipment at 400kV KHSTPP: ERLDC	8
2.5 PSDF funding for augmentation of Transmission Lines with HTLS: OPTCL .	9
2.6 Installation of new 50MVAR reactor at Jayanagar Substation: OPTCL.....	10
2.7 PSDF funding for establishment of Security Operations Center at DVC SLDC: DVC	10
2.8 Drawl mode at Maithon by IPCL post commissioning of 220 kV Maithon (ISTS) – Chalbalpur D/C line: ERLDC	11
2.9 Successful AMR data communication restoration in WB network: Powergrid ER-II	11
2.10 Re-configuration of LAN Network at respective state, for upgradation of Network in Layer3/Layer4: Powergrid ER-II	13
2.11 Update on Rajarhat GIS (POWERGRID) 400/220kV S/S: 2x500MVA and : ERLDC.....	14
2.12 Update on Patna Islanding scheme: ERPC.....	16
2.13 Frequency Response Performance (FRP) of Generators: ERLDC	19
2.14 Verification of Average Generation/Demand data for assessment of Frequency Response Obligation (FRO) for FY 2025-26: ERLDC	20
2.15 Generation target for FY 2025-26: ERPC	21
2.16 Self-Audit in compliance with clause 56(2) of IEGC: ERLDC.....	22
2.17 Shutdown request by DANS Energy.....	22
2.18 Periodic Testing of power system elements: ERPC.....	23
3. PART-C: ITEMS FOR UPDATE/FOLLOW-UP/INFORMATION	25
3.1. ER Grid performance during January 2025.....	25
3.2. Update on Reconductoring of ISTS lines under Eastern Region Expansion Scheme-44: ERPC	25
3.3. Update on Restoration of 132kV Rangit-Kurseong & 132kV Siliguri-Melli-Rangpo lines: ERLDC	27
3.4. Regarding Non-Submission of Forecasting Data from States: ERLDC	29
3.5. Non-Submission of FRC data in stipulated timeframe: ERLDC	31

3.6.	Mock Black Start: ERLDC	32
3.7.	Commissioning Status of ADMS: ERLDC	34
4.	PART-D: OPERATIONAL PLANNING.....	35
4.1.	Anticipated power supply position during March-2025	35
4.2.	Major Thermal Generating Units/Transmission Element outages/shutdown in ER Grid (as on as on 16-02-2025)	35
4.3.	Commissioning of new units and transmission elements in Eastern Grid in the month of January -2025.....	39
4.4.	UFR operation during the month of January 2025	41

EASTERN REGIONAL POWER COMMITTEE

MINUTES OF 224th OCC MEETING HELD ON 25.02.2025 (TUESDAY) AT 10:30 HRS

Member Secretary, ERPC chaired the 224th OCC meeting. On welcoming all the participants, he outlined the performance of ER grid during **January-2025** and highlighted the following points:

- ❖ In **January-2025**, energy consumption of ER was **14419 MU** which is **6% more than January-2024**.
- ❖ In **January-2025**, Peak demand met of ER was **24616 MW** which is **11.5% more than January-2024**.
- ❖ During **January-2025**, **76%** of time, the grid frequency was in IEGC Band (49.90Hz-50.05Hz).
- ❖ Thermal **PLF** of ER during **January-2025** was **79.4%**.
- ❖ Some thermal generating units were lauded for maintaining PLF more than **90%** during **January-2025** that are listed below:

Utility	Generating Stations	PLF %
WBPDCL	Bakreswar TPS	101
	Santaldih TPS	102
	Bandel TPS	95
NTPC	Darlipali STPS	100
IPP	Haldia TPP(HEL)	96
	Kamalanga TPS(GMR)	93
	Mahadev Prasad STPP(APRNL)	90

❖ **Coal stock position:**

☐ Coal stock position (As on 22.02.2025) is as follows:

SL.	Name of States/Power Stns.	% of Actual Stock vis-à-vis Normative Stock
1.	Jharkhand (TVNL)	97%
2.	Odisha/IBTPS	52%
3.	WBPDCL	30%(Min. Santaldih TPS -16 %, Max. Sagardighi TPS – 40 %)
4.	D.P.L. TPS	12%
5.	DVC	88%(Min. Mejia TPS-62%, Max Kodarma TPS - 134%)

6.	NTPC	97% (Max.Farakka TPS-144% & Nabinagar STPP-159%,Min.Talcher STPS - 60%)
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❖ **Transmission line (220 kV & above) commissioned during January-2025:**

- ✓ LILO Portion (10.5 km) of 400KV-BARH-BAKHTIYARPUR(BH)-1(QUAD MOOSE) (BSPTCL) first time charged on 31.01.2025.

❖ He further highlighted the following:

- ✓ On **20th Feb 2025** a grid disturbance occurred at **16:20 hrs** at 765/400kV Angul Station, 400kV GMR, 400kV JIPL Generating station.

Generation loss of around **1750 MW** occurred at GMR and JITPL due to loss of evacuation path. Inclement weather and cyclonic storm were reported during the event. Multiple CT faults were observed during the event and flashover marks on CT Junction box observed.

- ✓ Hon'ble Member(Technical), CERC held a discussion with ER states on 22.02.2025 in presence of ERPC & ERLDC wherein following key facets were stressed:

- ☐ Resource adequacy during the crunch period Apr-June 2025.The quantum of reserves should be maintained at a constant level to take care of any contingency.
- ☐ Implementation of MTDL(55%) in intra-state generators.
- ☐ Implementation of AGC in all ER state generators.

- ✓ Stressed on long-term planning i.r.o adequate and generation capacity and augmentation of transmission capability.

- ✓ All Eastern region DISCOMs were made aware of India's share allocation(state-wise) in Punatsangchu HEP of Bhutan for streamlined planning of intra-state demand portfolio.

- ❖ **ED,ERLDC** at the outset urged for comprehensive resource adequacy planning by all ER states in view of upcoming crunch demand period of Summer 2025. Further, timely and accurate demand forecasting by all states in compliance to IEGC was underscored as was also directed by CERC to all states in recent visit to ER.

1. PART-A: CONFIRMATION OF MINUTES

1.1. Confirmation of Minutes of 223rd OCC Meeting held on 24th January 2025 physically at Kolkata

The minutes of 223rd Operation Coordination Sub-Committee meeting held on 24.01.2025 was circulated vide letter dated 03.02.2025.

Members may confirm the minutes of 223rd OCC meeting.

Deliberation in the meeting

- ✦ *In line with observations received from NTPC, some alterations are hereby incorporated in the Minutes of 223rd OCC Meeting as detailed hereunder:*
- ***In Item no:B.2.1.: Disturbance at 400 kV NTPC Barh on 4th Jan 2025 at 19:23 Hrs:***
“NTPC submitted that Backup overcurrent protection shall be deployed with pickup above thermal limit having AND logic of VT supervision to avoid GT failure in case of any eventuality of similar nature.”
- ✦ *This modification shall form part and parcel of MOM of 223rd OCC circulated vide letter dated 03.02.2025.*
- ✦ *Members confirmed the Minutes of 223rd OCC with aforementioned modification.*

2. PART-B: ITEMS FOR DISCUSSION

2.1 SOP on Shifting of Transmission lines for other infrastructure projects: ERPC

- ✓ Several infrastructure projects of National Highways Authority of India (NHAI), Indian Railways, Airports Authorities, Border Roads Organization (BRO), Irrigation Departments, etc. are going on in various parts of the Country and many times, construction/development projects like roads, railways, airports, mines, flood banks/dam etc. come across existing/under construction transmission lines in their route alignment, leading to the need for shifting of such lines for construction of these projects.
- ✓ These projects often intersect with existing or under-construction transmission lines, necessitating their relocation to facilitate project implementation, However, inadequate coordination between Project Developers and owners of transmission lines - Transmission Service Providers/ Transmission licensees - frequently compromises the safety of transmission lines during construction and leads to significant delays in project completion.
- ✓ To address these challenges, ensure compliance with existing regulator' provisions issued by the Central Electricity Authority (CEA) and other authorities such as the Ministry of Environment and Forests, Defence, Airport Authority, NHAI, BRO, Indian Railways and ensure smooth coordination between the Infrastructure Developers & transmission line Owners, a Standard Operating Procedure (SOP) for shifting of Transmission lines by various Project Developers had been notified by CEA dated 10.03.2023.
- ✓ During the processing of various proposals of diversion of transmission lines involving works by other Project Developers, it has been observed by the Ministry of Power that there is a need to bring greater clarity in the definition/ parameters for considering a project as a project of National Importance or serving wider public interest.
- ✓ In view of simplifying and further streamlining the said SOP, Ministry of Power vide its letter dated 07.01.2025 conveyed approval of the Hon'ble Minister of Power for "Revised

Standard Operating Procedure for shifting of Transmission line for other infrastructure projects

□ Details of Projects of National importance or Serving Wider Public interest

(i) Projects monitored on the portals of Department for Promotion of Industry and Internal Trade (DPIIT) (Project Monitoring Group) or Ministry of Statistics and Program Implementation (MoSPI) shall be considered as projects of National Importance.

(ii) The projects recommended by Central Ministries/ Departments concerned or State Governments shall be considered as projects serving wider public interest.

(iii) Real Estate projects in accordance with the notified master plans or the layout plans approved by the local area authorities and recommended by the State Governments shall also be considered as projects serving wider public interest.

(iv) Accordingly, the attached Standard Operating Procedure (SOP) is issued for the relocation of Inter-State Transmission System (ISTS) lines during the development of infrastructure projects.

(v) This SOP shall be applicable only for cases where the Project Developers provide alternate Right of way (RoW) for shifting and not applicable for multi-circuiting on adjoining lines in the vicinity.

(vi) COO, CTUIL may refer cases, not covered under this SOP, to Ministry of Power, through Central Electricity Authority.

Details of SOP enclosed at **Annex-B.2.1.**

This is for information to and necessary compliance by all ISTS licensees along with State Transmission utilities.

Member may discuss.

Deliberation in the meeting

*A brief presentation (**Annex B.2.1**) was delivered by ERPC representative outlining the salient aspects of the revised SOP for shifting transmission lines by licensees that was recently circulated by Ministry of Power, Govt. of India.*

OCC Decision

*OCC noted and advised all transmission utilities (states as well as ISTS licensees including private licensees) to abide by the revised SOP(**Annex B.2.1**) i.r.o shifting the transmission lines for other infrastructure projects of national importance or wider public interest.*

2.2 Grid Disturbance at 765/400kV Angul S/S, 400kV GMR and 400kV JIPL: ERPC

- ✓ A disturbance occurred at 16:20 Hrs on **20.02.2025** at 765/400 kV & 400 kV Angul, GMR, JIPL S/s. 765 kV Bus-1&2, 400 kV Bus-1&2 at Angul S/S tripped and generation loss of around **1750 MW** occurred at **GMR** and **JITPL** due to loss of evacuation path.
- ✓ Inclement weather and cyclonic storm were reported during the event. Multiple CT faults were observed during the event and flashover marks on CT Junction box observed.
- ✓ Total duration of outage: **1 Hr and 37 Minutes**

- ✓ Due to inclement weather with cyclonic storm at 765/400kV Angul Station, 400kV Bus 1 & 2, 765 kV Bus 1 & 2 along with all transmission element emanating from Angul station (Except 400kV Bolangir feeder, 765kV Jharsuguda Feeder 2, 765/400kV 1500 MVA ICT 1 & 2) tripped. Tripping of 400kV Angul JIPL D/C line led to loss of evacuation path for JIPL Generating station. Due to this, both the running units (Unit 1 & 2) of JIPL got tripped at 16:20 Hrs. Similarly tripping of 400kV Angul GMR D/C line caused tripping of Unit #1 & 2 at 400kV GMR generating station.

❖ Following are the key observations:

- Few CTs where faults occurred are common in both incidents.
- It is gathered that only SF6 filled CTs are getting affected due to lightning. Oil filled CTs are not impacted.
- It is suspected that flashover is occurring at junction box due to overvoltage induced by lightning strike in both the events. After the past incident on 12.10.2023 it was recommended in 130th PCC minutes for earthing audit and DSLP (Direct Stroke Lightning Protection) study, which was carried out and as intimated recommended steps were carried out. However, same nature of incident occurred again.
- ✓ One past incident of same nature also occurred at **765/400 kV Angul S/S** on 12.10.2023 where due to lightning, multiple faults occurred at various CTs which led to bus tripping at 765 kV and 400 kV.

Flash report attached at **Annex-B.2.2**

Powergrid Odisha may explain. Members may discuss.

Deliberation in the meeting

Powergrid Odisha presented a detailed report (**Annex B.2.2**) on the Grid disturbance at 765 kV Angul S/S, encompassing the sequence of events along with preliminary findings.

Powergrid Odisha further submitted:

- ✓ Internal flashover was observed in the SF₆ filled CTs while oil filled CTs remained intact. The failed CTs were of the same make & type as that observed in the previous incident(12.10.2023).
- ✓ The primary cause of CT failure is that somehow lightning surge is entering the substation and travelling to the CT's, leading to CT failure, insulation breakdown, and flashover of CTs, so DLSP (Direct lightning stroke Protection) to be further reviewed.
- ✓ After the past incident in October 2023 in which 4 Nos of CTs failed, out of which they replaced two no's of CTs with Oil filled and two with SF6 filled CT and in recent event also total 6 Nos of SF6 CTs failed in which 2 Nos of CT which were earlier replaced with SF6 failed.
- ✓ SF6 CTs are more susceptible to failure during lightning surges due to their different capacitance compared to oil-filled CTs. While oil-filled CTs exhibit greater resilience to lightning surges, they present a higher risk of bursting in case of failure and causing significant damage to adjacent equipment.
- ✓ Concerned OEM of the CTs has been called in to carry out Root cause analysis(RCA). Report will be submitted within a month.

- ✓ Previously a third party earthing audit was carried out following the incident in October 2023 and some findings regarding earthing pointed out by an external agency have been duly addressed. The recommended actions, such as strengthening earth pits and raising riser heights, were subsequently implemented.
- ✓ The disturbance took place despite DSLP(Direct Stroke Lightning Protection) being in place at Angul S/S.
- ✓ As suggested by Powergrid Corporate Engineering team, measures have been taken to strengthen DSLP(Direct Stroke Lightning Protection) with deployment of additional earthwires.
- ✓ Study on proper discharge of lightning impulse to the ground shall be carried out.

ERLDC raised the following observations:

- ✓ Similar incidents involving multiple faults in Current Transformers (CTs) due to lightning occurred at the 765/400 kV Angul Substation (S/S) on October 12, 2023. Where similar faults resulted in bus tripping at both 765 kV and 400 kV levels. Notably, some of the CTs affected in the previous incident were also involved in the current event.
- ✓ It is gathered that only SF6 filled CTs have been affected due to lightning, while oil filled CTs are unaffected.
- ✓ An investigation is required to determine if the problem is confined to a specific batch of CTs. Furthermore, the root cause analysis (RCA) of the failed CTs from the previous incident is still outstanding. This analysis must be completed for the current event as well, and the RCA report should be shared as soon as possible.
- ✓ It is suspected that flashover is occurring at junction box due to overvoltage induced by lightning strike in both the events. After the last event, an Earthing audit was carried out and the Compliance of recommendations to be submitted as the same nature of event occurred again.
- ✓ As GMR and JITPL are connected to the same Dia, a simultaneous outage of both buses will result in generation loss. To minimize this risk of generation loss in future, possibility of shifting generator bays to a future bay within the Dia connected to the 765/400 kV ICTs or any other lines needs to be explored.

OCC Decision

- OCC opined that occurrence of similar nature of fault being repetitive at 765 kV Angul S/S, the same may be attributed to certain design deficiency in the system.
- Powergrid Odisha was advised to carry out root cause analysis, thereby pin-pointing the exact cause for failure of DSLP(Direct Stroke Lightning Protection) at Angul S/S and submit the findings in next OCC.
- Powergrid Odisha was also advised to take up with concerned OEM of the CTs and ascertain the reasons of failure along with RCA report in next OCC. Based on the outcome, Powergrid may take suitable decision.
- Implementation status of previous third party earthing audit of the entire Angul substation should be furnished by Powergrid Odisha in the next OCC.
- Feasibility of shifting of GMR /JITPL Dia to be explored by Powergrid Odisha at Angul S/S for taking up the issue for further implementation.
- OCC recommended for constitution of a committee by ERLDC comprising members from ERLDC, Powergrid and CPRI for detailed analysis of the causes of CT failure during Grid

disturbance at 765 kV Angul(PG) S/S. A comprehensive report delineating the reasons and preventive measures needs to be submitted within 3 months for review in OCC forum.

2.3 Bus split operationalization at NTPC Kahalgaon: ERPC

As decided in 219th OCC Meeting, a committee comprising of members from ERPC and ERLDC visited NTPC Kahalgaon on 17-10-2024 to assess the status of Bus splitting at 400 kV level and way forward for operationalization of 400 KV Bus sectionalizer.

Following works need to be done to complete the installation of ICT 3 & 4:

1. Determination of underground cable conduit path for 400/132 kV ICT-3, 4 and 5 allocated for stage 2 supply.
2. Excavating the existing cable and relaying from Stage-1 132kV to New Stage-2 132 kV switchyard, where ICT 3 & 4 will be connected.
3. Laying of additional 22.8 ckt. km control cable for STs.
4. Jumpering of ICTs in 132kV & 400kV level.
5. Bay equipment testing.
 - NTPC apprised that determination of underground power cables is one of the major challenges to proceed further with laying of cables between two 132kV switchyards. The tentative time to complete the ICT commissioning is **25th May 2025**.
 - Meanwhile in view of increased fault level of NTPC Kahalgaon and to facilitate interim arrangement of standby ISTS connectivity to Godda Thermal Power project of M/s Adani Power (Jharkhand) Ltd. (APJL) with Indian grid, Bus splitting at 400KV Kahalgaon needs to be done on priority.

222nd OCC Decision

In view of continued operation of 400 kV bus in synchronised mode at NTPC Kahalgaon at high fault level, OCC expressed serious concern over the lack of desired progress in 132 kV power cable laying and thereby suggested participation of representative from concerned site i.e NTPC Kahalgaon in next OCC meeting for clarity i.r.o progress in bus splitting activities at the site.

As per the latest deliberation in 53rd TCC meeting:

- ❖ RED, NTPC(ER-I) informed:
 - ✓ Out of total 18 jobs involved in bus splitting, 14 jobs have already been completed.
 - ✓ Power cable and control cable laying is under progress. Laying of power cable is getting delayed due to obstructions encountered in form of live CW ducts, fire hydrants, etc in a fully operational thermal power plant.
 - ✓ Bus split at 400 kV Kahalgaon to be made operational by June 2025.
- ❖ As per information received from Adani Power, standby ISTS connectivity to Godda thermal power project is to be completed tentatively by July 2025.

Latest status as furnished by NTPC enclosed at **Annex-B.2.3**

NTPC may confirm timeline and feasibility of Bus splitting with current configuration. Member may discuss.

Deliberation in the meeting

NTPC submitted:

- ✓ A Gantt chart was shared detailing the timeline of activities and completion target by June 2025.
- ✓ Excavation process for laying of 132 kV Power cables has already started. Laying of Power cables shall commence from First week of March 2025. Presence of CW ducts and Fire Hydrant pipes in the path of the cable has delayed progress.
- ✓ Control cables for ICT charging are being sourced from other NTPC projects.

OCC Decision

- OCC urged NTPC to strictly adhere to the committed timeline for bus splitting at Kahalgaon, i.e June 2025.
- NTPC ER-I Headquarters was advised by OCC to facilitate the availability of control cables at NTPC Kahalgaon.
- NTPC was advised to submit fortnightly progress report to ERPC/ERLDC with timeline of all intermediate activities (target v/s progress achieved).

2.4 Reliability and Healthiness of equipment at 400kV KHSTPP: ERLDC

- ✓ Emanating lines from KHSTPP (NTPC) which are equipped with BHEL-manufactured circuit breakers have exceeded 25 years of service. These aging assets have deteriorated significantly due to hydraulic mechanism issues and increased SF6 leakage.
- ✓ Furthermore, several isolators located at KHSTPP(NTPC) are stuck-up and inoperable. In addition to these challenges, KHSTPP has been experiencing extended outages of multiple critical elements. The details of these outages are as follows:

Name of the Elements	Remarks
400kV-Kahalgaon- Barh-2	Out of service since 07.12.24
400kV tie bay of FSTPP-1 & Barh-2	Out of service since 18.10.24
400kV Tie Bay of Durgapur-2 & Future	Out of service since 06.07.24
400kV Main Bay of Banka-1	Out of service since 24.02.24
400kV Main Bay of Kahalgaon-Lakhisarai-1	Reliability of the BHEL make CB has significantly deteriorated due to issues with their hydraulic mechanisms and increased SF6 leakage. These CBs are being obsolete models over 25 years old. (4 nos. of CB)
400kV Tie Bay of 400/132kV ICT-2 and Kahalgaon-Lakhisarai-1	
400kV Main Bay of Kahalgaon-Lakhisarai-2	
400kV Tie Bay of Kahalgaon-Durgapur-1 and Kahalgaon-Lakhisarai-1	
Bus-3 side Isolator of Bus-1 & 3 sectionaliser	Isolators are in stuck-up condition (9 nos. of isolators)
Bus-1 side Isolator of Bus-1 & 3 sectionaliser	
Bus-4 side Isolator of Bus-2 & 4 sectionaliser	
400kV Main Bay of Kahalgaon-Banka-2	
400kV Tie Bay of Kahalgaon-Banka-2	

- ✓ With the approaching summer season, power demand will increase significantly and power flow in this corridor during solar hours will surge due to high solar injection from solar rich Northern region & high space cooling load in West Bengal.

- ✓ Being a critical substation in the high-power flow corridor, the healthiness of all equipment needs to be ensured to maintain a stable electricity supply.

Moreover, it is observed that other NTPC stations also lacking in maintenance activity over past few years.

ERLDC may explain. NTPC may update. Members may discuss.

Deliberation in the meeting

- ❖ ERLDC presented details of maintenance carried out in NTPC generating stations w.r.t Bus, ICTs and Line Bay over last four years. (**Annex B.2.4**). It was informed that maintenance of Bus & ICTs is irregular at Barh, Darlipalli, Farakka and Nabinagar generating stations while line bay maintenance is regular only at Barh, Farakka and Kahalgaon stations.
- ❖ ERLDC also raised concerns in the forum regarding the multiple circuit breaker (CB) outages and stuck isolators at Kahalgaon STPP. These incidents have not only reduced the reliability of Kahalgaon STPP but have also impacted the reliability of neighbouring generating stations, including Farakka and Barh, as well as the overall stability of this critical power corridor.
- ❖ NTPC Kahalgaon STPP informed the forum that a design-related issue with Siemens-manufactured isolators is the cause of the stuck isolators. They confirmed that a replacement plan is already in place.
Kahalgaon STPP was requested by ERLDC to share the detailed work plan and progress updates for these replacements.

OCC Decision

- OCC opined that lack of proper maintenance at NTPC generating stations may potentially cause forced outage of switchyard equipment and hence generating unit(s) .
- NTPC was advised to be vigilant in ensuring proper and timely maintenance activities at switchyards of all their generating stations.
- NTPC was advised to plan for maintenance activities in coordination with ERLDC.
- OCC advised ERLDC to regularly monitor the status of maintenance in all generating stations of ER and highlight the deficiencies in OCC forum.

2.5 PSDF funding for augmentation of Transmission Lines with HTLS: OPTCL

Two transmission lines of OPTCL require augmentation to HTLS conductors. These lines are of strategic importance, carrying bulk power to vital load centers, any disruption or failure in these lines can seriously affect the load centers. The concerned lines are: -

- 400kV IB – Lapanga D/c line
- 220kV Budhipadar – Lapanga – Katapali D/c line.

OPTCL has requested for the funding of the project under PSDF. A detailed Load Flow Analysis report for the above lines is attached as **Annex B.2.5**.

OPTCL may explain. Members may discuss.

Deliberation in the meeting

OPTCL presented the load flow study and explained the necessity of reconductoring w.r.t 400kV IB – Lapanga D/c and 220kV Budhipadar – Lapanga – Katapali D/c lines.

OCC Decision

- *Considering persistent high loading in above mentioned transmission corridors, OCC technically consented to the proposal of HTLS reconductoring and OPTCL may approach PSDF for funding.*

2.6 Installation of new 50MVAR reactor at Jayanagar Substation: OPTCL

- ✓ *The 220kV/132kV Jayanagar Substation is experiencing overvoltage issues in the 220kV switchyard during the winter off peak condition. To address this issue a 50MVAR reactor may be installed to maintain the voltage within 1.03pu.*
- ✓ *OPTCL has requested for funding of this plan with the PSDF fund. A detailed report for the above plan has been attached in **Annex B.2.6.***

OPTCL may explain. Members may discuss.

Deliberation in the meeting

OPTCL informed that high voltage is experienced at 220 kV Jaynagar substation in winter(off peak season) necessitating installation of 50 MVAR reactor for maintaining bus voltage within permissible limits(+/- 3%).

OCC Decision

- *OCC technically consented to the proposal and OPTCL may approach PSDF for funding.*

2.7 PSDF funding for establishment of Security Operations Center at DVC SLDC: DVC

As per the **Information Technology (Information Security Practices and Procedures for Protected Systems) Rules 2018**: -

- ✓ *An organisation having “Protected System” shall establish a Cyber Security Operation Centre (C-SOC) using tools and technologies to implement preventive, detective and corrective controls to secure against advanced and emerging cyber threats.*
- ✓ *In addition, C-SOC is to be utilised for identifying unauthorized access to “Protected System”, and unusual and malicious activities on the “Protected System”, by analysing the logs on regular basis.*
- *Accordingly, a plan has been drafted by DVC establish Security Operation Centre at DVC LDC, 7th floor, DVC TOWERS, Damodar Valley Corporation, CIT Rd, Ultadanga, Kolkata, West Bengal 700054 to ensure compliance of the directives of **Information Technology (Information Security Practices and Procedures for Protected Systems) Rules 2018** & to ensure proper monitoring & security of CII of DVC LDC.*
- *In this regard, a DPR has been prepared by DVC for the Establishment of SOC at DVC LDC for the grant of funding from PSDF under category 5.1(c), as mentioned in **Annex B.2.7.***

DVC may update. Members may discuss.

Deliberation in the meeting

DVC submitted DPR with tentative cost for implementing SOC at DVC LDC in compliance to Information Technology rules 2018.

OCC Decision

- *OCC technically consented to the proposal and DVC may approach PSDF for funding.*
- *OCC also advised other ER states except Bihar to expedite SOC implementation.*

2.8 Drawl mode at Maithon by IPCL post commissioning of 220 kV Maithon (ISTS) – Chalbalpur D/C line: ERLDC

- ✓ IPCL has been granted 100 MW of General Network Access (GNA) under transition, as per Section 37.2 of the GNA Regulations. The connectivity is granted through the 220 kV Maithon (ISTS) – Chalbalpur D/C line, which is to be implemented by IPCL. The GNA has been made effective from 1st October 2024; however, the associated transmission line is yet to be commissioned the same line is expected by December 2026.
- ✓ Since the 220 kV Maithon (ISTS) – Chalbalpur D/C line are not yet operational, it is understood that presently the load is the said licensed area being supplied entirely through the STU system.
- ✓ Despite multiple communications, no response has been received from IPCL regarding the operational mode of power drawal post-commissioning of the line.
- ✓ It is expected that post commissioning of 220 kV Maithon (ISTS) – Chalbalpur D/C, IPCL will draw the power in radial mode. The licensed area connected with CTU and STU systems will not be connected.

It may be noted that:

- Post commissioning of the dedicated line, if IPCL plans to connect both STU and CTU systems in a meshed configuration, prior intimation to be given to ERLDC before implementing the change.
- Once the STU and CTU systems are synchronously connected, NLDC shall discontinue the computation of separate calculation of IPCL's CTU system transmission charges, aligning with the treatment given to Torrent Power Limited.

ERLDC may explain. Members may discuss.

Deliberation in the meeting

ERLDC submitted :

- ✓ *IPCL has been granted 100 MW GNA under, effective from 1st October 2024, through the 220 kV Maithon (ISTS) – Chalbalpur D/C line, which is yet to be commissioned (expected by December 2026).*
- ✓ *No response has been received from IPCL regarding the operational mode post-commissioning of this line. It is expected that IPCL will draw power in radial mode, and if a meshed connection with STU and CTU systems is planned, prior intimation must be given to ERLDC.*
- ✓ *Post synchronization of both systems, separate CTU transmission charge computation for IPCL will be discontinued.*

OCC Decision

OCC opined that the issue having commercial implications is being referred to CCM .

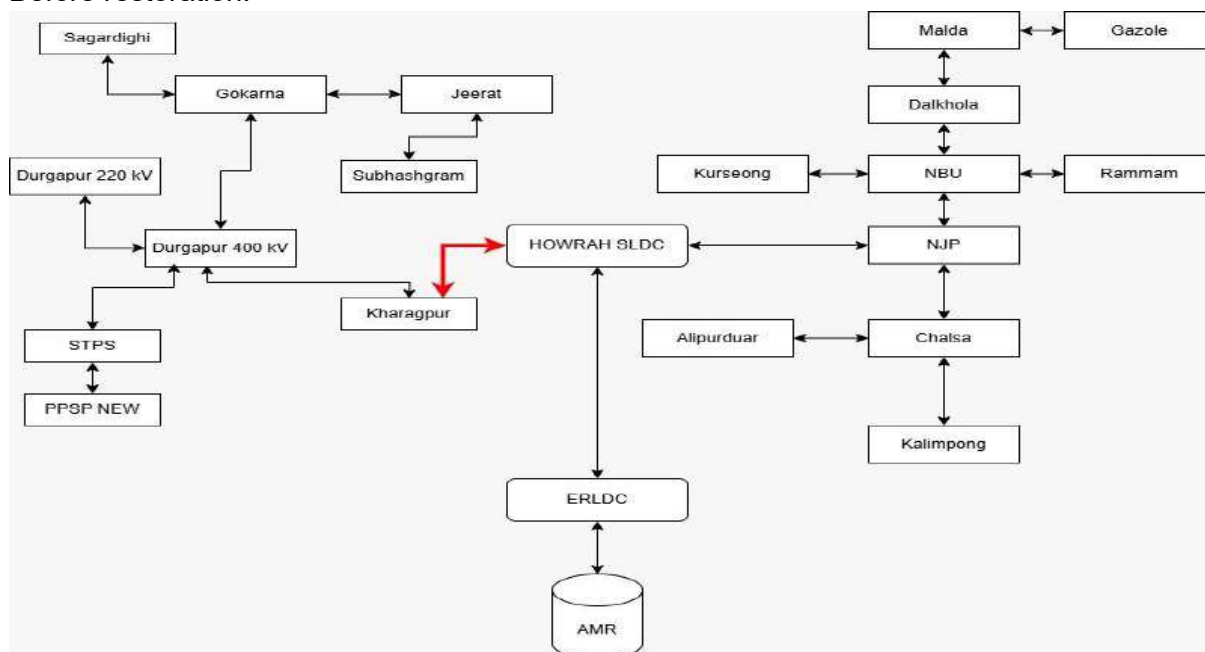
2.9 Successful AMR data communication restoration in WB network: Powergrid ER-II

- ✓ In West Bengal, total number of substations connected with AMR system is: 22. All these locations were successfully communication with AMR system and SEM data communication was OK.
- ✓ In July 2024, a single port was provided by network team, for all WB locations at ERLDC data center. Previously, there were 2 ports that were shared with other locations, such as DVC, PG, central sector etc. (the changes were done after the data broadcast issue

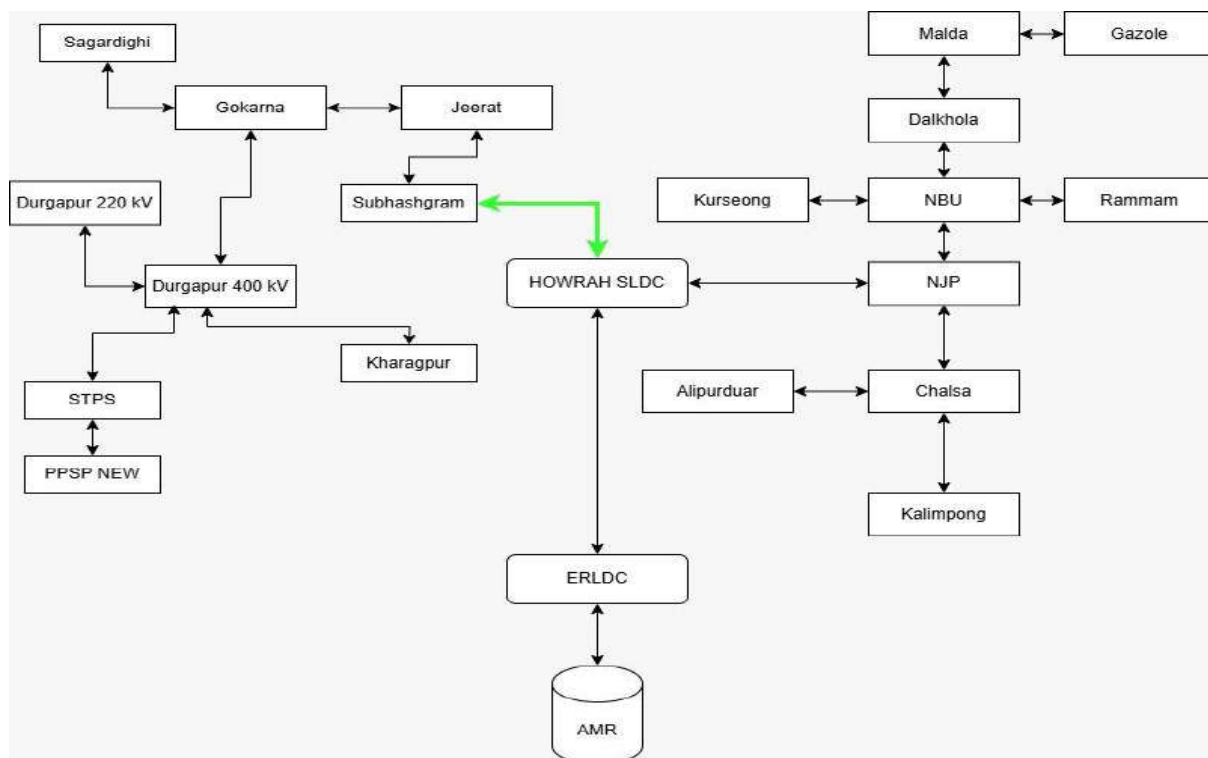
observed). Since the change in configuration, it was seen that for 8 WB locations (South zone i.e: PPSP, Bidhannagar, Subhasgram, New Town, Bidhannagar, Jeerat, Santaldihi, Sagardighi) that even though ping was available, AMR data transfer through secure FTP was not happening. AMR system has same setup in other locations across entire ER and no issues were noticed in these places.

- ✓ Multiple tests (ping test with load, FTP test, Firewall rule check etc.) were performed at ERLDC data center and Substation end, however, root cause of issue was not identified.
- ✓ In 16th TEST meeting it was mutually decided that another round of comprehensive testing of WB LAN will be conducted. On 4th February 2025, 3 teams were deployed at 3 separate locations. One team at a chosen substation (from the 8 affected ones), Subhashgram (WB) in this case, one at WB SLDC (Howrah) and one at ERLDC data center. The communication channel from SLDC to ERLDC was found to be OK and ftp was happening. The network connection from substation to SLDC was not working and after rigorous testing of each individual node, the issue was narrowed down to Kharagpur (WB) junction where the VLAN for all the affected locations branched out from (refer to image). Then, as per the suggestion of AMR network team, the connection from Kharagpur (WB) was re-routed to Subhashgram (WB).
- ✓ After the re-routing of the virtual network, it was seen that ftp was once again working. AMR data transfer from all substations to ERLDC data center was working correctly. Therefore, it can be ascertained that AMR system was not the root cause of the issue and that it was in the network end, particularly in the configuration of the locations of South WB VLAN. Referring to the image below, it can be seen that since July 2024 the VLAN connection path was routed through Kharagpur (WB) for the affected south WB locations (marked in red) and is the sole reason for non-communication. Now that route has been changed and the VLAN is configured to be routed through Subhashgram (WB).

Before restoration:



After restoration:



All WB nodes are ready for communication to AMR wherever LAN connectivity is available.

Powergrid ER-II may explain. Members may discuss.

Deliberation in the meeting

Powergrid informed:

- ✓ On shifting the VLAN from Kharagpur (WB) to Subhasgram(WB), uninterrupted AMR data transfer through secure FTP in entire WB network was ensured.
- ✓ Except DVC and now West Bengal, none of the ER SLDCs is having a single node or dropping point for AMR communication.

OCC decision

- OCC advised SLDCs of Bihar, Jharkhand, Sikkim and Odisha to coordinate with respective communication teams and arrange for single dropping point i.r.o AMR data communication to ERLDC.
- OCC acknowledged Powergrid for resolving the persistent issue of AMR communication in WB network.

2.10 Re-configuration of LAN Network at respective state, for upgradation of Network in Layer3/Layer4: Powergrid ER-II

The upgradation of existing AMR Network project to Layer3/Layer4 has been approved in 52nd TCC and 219th OCC meeting. LOA is already placed to M/S TCS on 27.01.2025 and targeted completion schedule is July-2025.

As decided and agreed by all the states, following are the areas which the respective state has to ensure for execution of the project in proper time: -

- All respective state/utilities must drop all the AMR communication points to a single node/dropping point at SLDC. There shouldn't be multiple dropping points of AMR communication network channel for a single state. This activity must be done before the router installation.
- All state/utilities must provide the Rack space and power supply for installation the Routers.

This change in network configuration from all states needs to be completed within 20th March 25 and the confirmation of the same should be shared with PGCIL/ERPC/ERLDC. Based on the confirmation, the site survey and the other pre testing will be performed before the installation of Routers and final activities.

Powergrid ER-II may explain. Members may discuss.

Deliberation in the meeting

Powergrid ER-II updated the latest status as:

LOA has been placed to M/S TCS and and AMR network upgradation is scheduled to be completed by July 2025.

OCC Decision

- *OCC advised all SLDCs to facilitate single node/dropping point for all AMR communication points for seamless upgradation of ER AMR network to Layer 3/ Layer 4 and thereby averting possibility of data broadcast. Additionally, rack space with power supply must be arranged by the states for router installation.*
- *All these activities need to be completed by 25th March 2025 and confirmation should be intimated to Powergrid for carrying out site survey.*

2.11 Update on Rajarhat GIS (POWERGRID) 400/220kV S/S: 2x500MVA and : ERLDC

- ♦ During the deliberations in the 222nd OCC meeting held on 23.12.24, West Bengal SLDC representative expressed deep concern regarding a potential power crisis at Rajarhat (PG) in 2025-2026, based on the current loading pattern in and around the Kolkata area. It was also emphasized the need to prioritize the installation of a 3rd 400/220KV,500MVA ICT at Rajarhat (PG) with the same urgency as Subhasgram (PG) to prevent a recurrence of similar critical situations in the future.
 - ♦ It was further highlighted that if the proposed 3rd ICT is not operational by the summer of 2026, severe congestion is likely to affect the ICTs at Rajarhat (PG). Additionally, it was noted that the state assembly elections are expected to take place in 2026, adding to the significance of addressing this issue promptly.
 - ♦ Powergrid ER-II updated in the meeting that tender for procurement of the 3rd ICT has been annulled twice and currently re-tendering is under progress. Bid opening is scheduled tentatively in Feb 2025, thereafter, commissioning of the ICT will take 18 months. It is expected to be commissioned by end of 2026 or first half of 2027.
 - ♦ In view of the above, an alternative arrangement may be thought off to ensure 3rd ICT at Rajarhat before 2026 Summer.
- The need for augmenting the transformation capacity at Subhasgram, Kolkata, was first discussed during the **45th TCC** meeting on **06.08.2022** in Siliguri, West Bengal.
- Subsequently, considering the urgency of augmenting ICT capacity to meet the increasing

demand in the Kolkata area, it was decided during the 50th TCC/ERPC meeting on 10.08.2023 at Lonavala to utilize the regional spare 500 MVA ICT located at Maithon as 7th ICT at Subhasgram. This ICT was transported and commissioned in June 2024, thereby enhancing the transformation capacity for the Kolkata area.

As per update in **53rd TCC** meeting

❖ Powergrid ER-II updated:

- ✓ The 6th ICT at Subhasgram(PG) can be made available from Maithon(PG) as discussed in recent bilateral meeting with WBSETCL but transportation from Maithon(PG) would be extremely challenging.
- ✓ All civil and electrical works for 6th ICT at Subhasgram(PG) to be tentatively completed by December 2025.
- ✓ Tender for procurement of the 3rd ICT at Rajarhat(PG) had been annulled twice. Presently re-tendering under progress whose NIT will be floated within 7 days. LOA shall be placed by August 2025.
- ✓ The 3rd ICT can be commissioned at Rajarhat (PG) by August 2026.

❖ WB SLDC submitted :

- ✓ Raised serious concern over the inordinate delay, especially in initiating the re-tendering process by Powergrid and consequently lapsing crucial 19 months in between.
- ✓ Despite continuous pursuance in several OCC meetings, no significant headway could have been achieved in this regard.
- ✓ It was highlighted that while Summer 2025 will be difficult but Summer 2026 will definitely encounter a power supply crisis i.r.o Kolkata in absence of 3rd ICT at Rajarhat(PG).
- ✓ Regular monitoring of the progress was requested.
- ✓ Interim solution of load management in upcoming Summer with existing resources was requested.

Letter in this regard from NLDC is attached as **Annex B.2.11**

ERLDC may explain. POWERGRID ER-II may update. Members may discuss.

Deliberation in the meeting

POWERGRID ER-II updated:

- ✓ *The GIS package at Rajarhat(PG) is yet to be finalized.*
- ✓ *The 3rd ICT(500 MVA) is expected at Rajarhat(PG) by December 2026.*
- ✓ *As per MOM of the meeting chaired by Power Secretary (Govt of WB) dated 20.02.2025, the 500 MVA transformer has been planned for transportation from ER spare pool at Maithon (PG) to Subhasgram (PG) tentatively in April 2025. Necessary administrative clearance & assistance shall be required to facilitate challenging transportation process.*

OCC Decision

- *OCC noted and advised Powergrid to abide by the submitted timeline of 500 MVA ICT commissioning i.r.o both Rajarhat(PG) and Subhasgram (PG) .*
- *Powergrid was advised to submit to ERPC Secretariat: Gantt chart with timeline of each activity to be carried out as well as fortnightly progress report on ICT commissioning at Rajarhat and Subhasgram substations.*

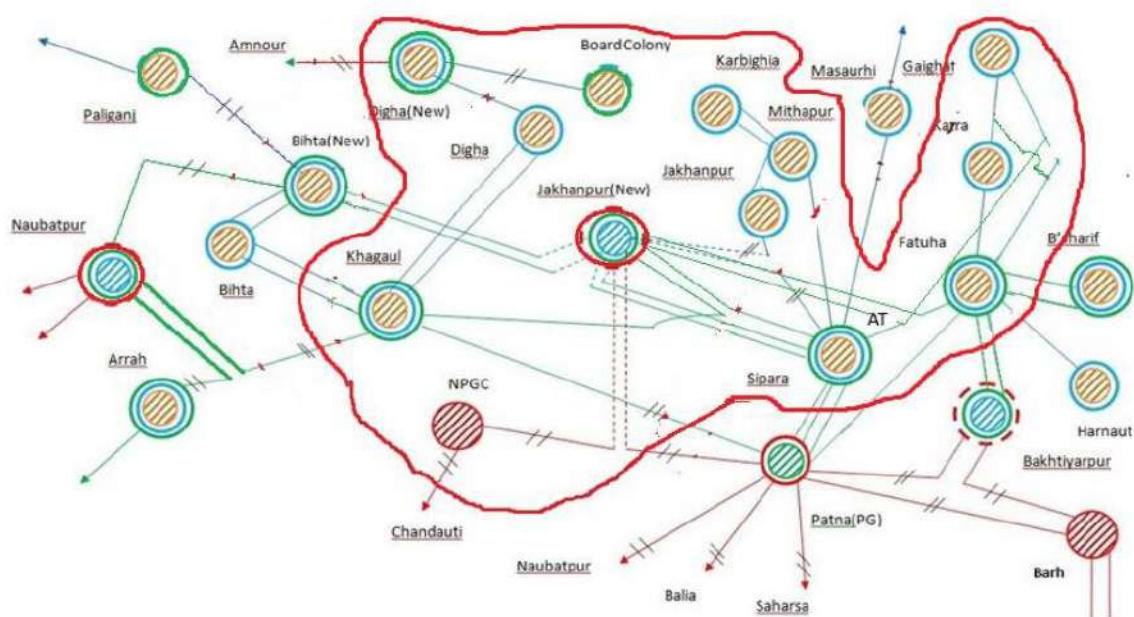
2.12 Update on Patna Islanding scheme: ERPC

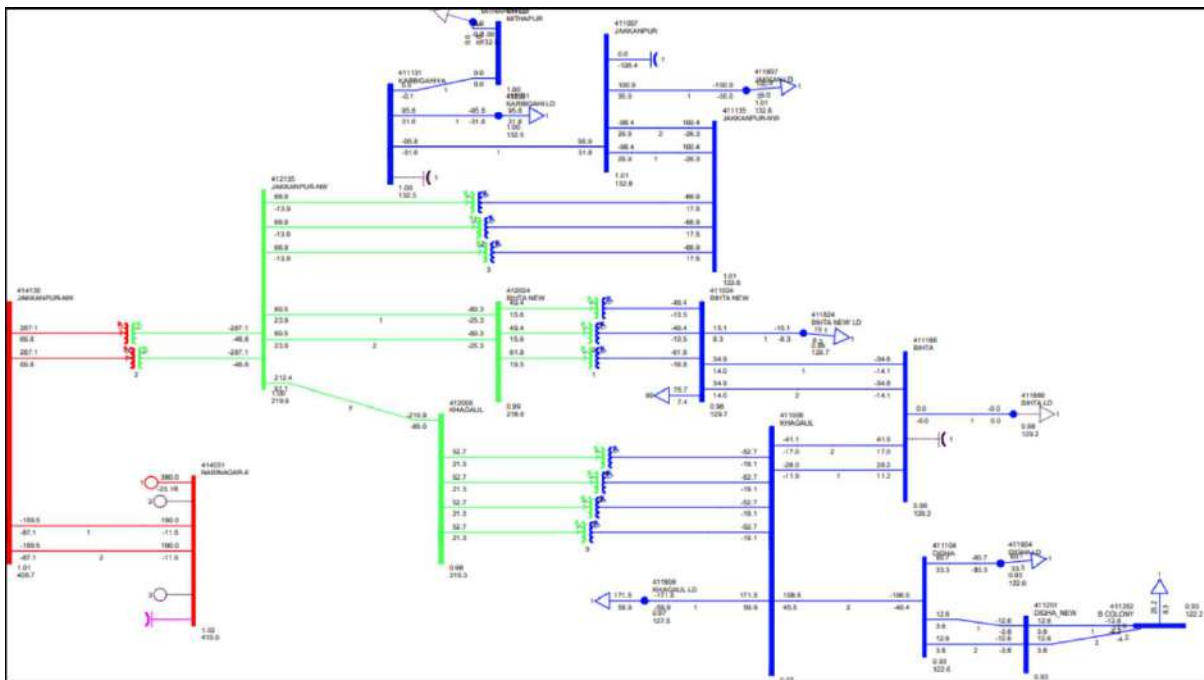
The Patna islanding scheme would be formed with Units of NPGCL along with loads of Patna city.

NTPC was entrusted for carrying out study of NPGC units and M/S Solvinia had submitted report on study of islanding scheme dated 08th May 2024. Thereafter based on comments received from ERLDC, replies were submitted by M/S Solvinia. NTPC had communicated the report to all concerned including SLDC Bihar.

Some further tests needed could not be carried out due to non-receipt of relevant data from Bihar.

- The proposed Patna islanding scheme aims to isolate one running unit of NPGC (660 MW) with pre-identified load of Patna city and nearby areas. After isolation of selected loads and NPGC through the identified network, run the island in islanded mode to cater the city load and to extend start-up supply to generating stations in adjoining area to facilitate early restoration.
- Patna city and nearby loads will be islanded with one of the running units of NPGC (660 MW). NPGC is connected to the grid through 400 kV NPGC-Jakkanpur D/c and 400 kV NPGC Gaya D/c lines. For the islanding 400 kV NPGC-Jakkanpur D/c and at Jakkanpur through 400/220 kV ICTs, pre-deintified 220 kV feeders will be selected which will be isolated to confirm the islanding of the Patna loads from the rest of the grid with one unit of NPGC.





Islanding Logic(proposed by ERLDC):

As demand of identified feeders may increase/decrease with time, to maximize chance of survival, it is necessary to have a central logic system which will monitor load and generation balance and will trip feeders prior to islanding if frequency reaches below a certain point.

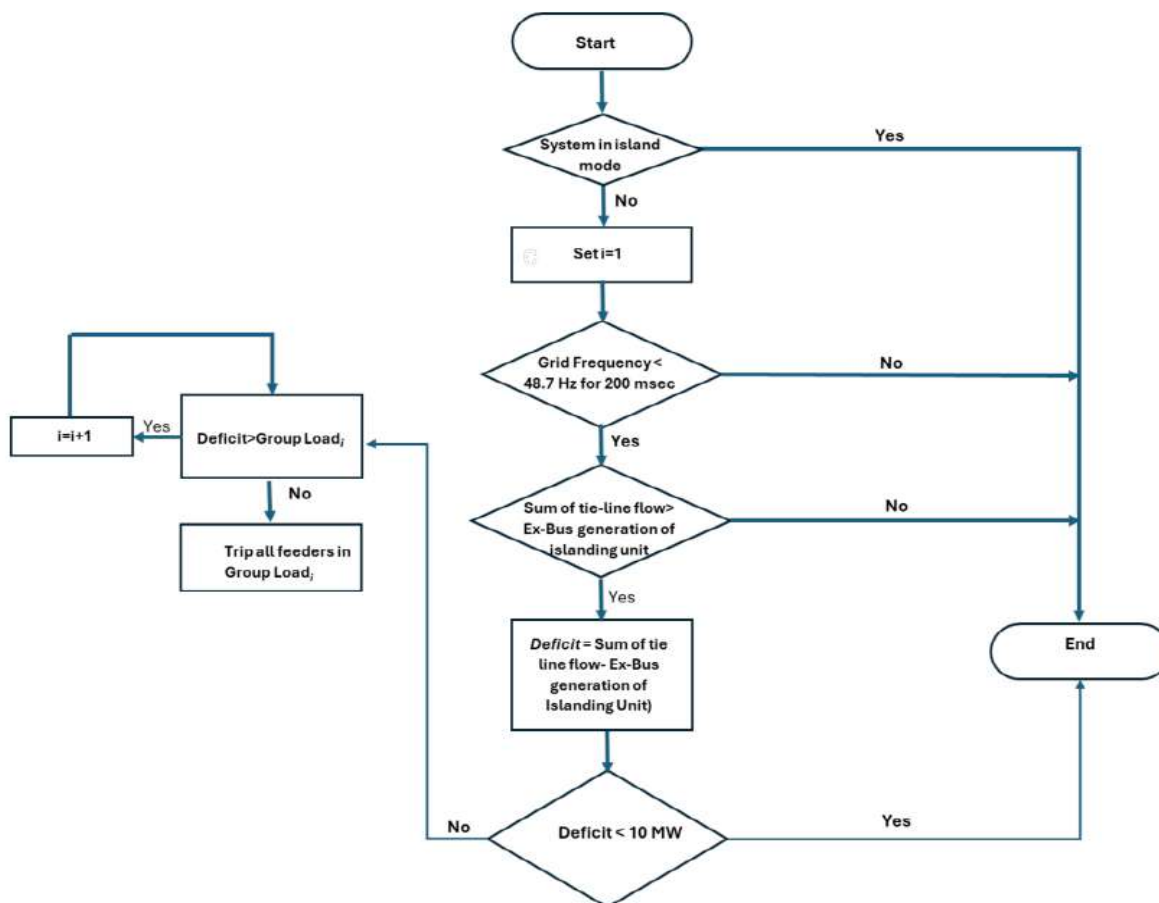
- **Pre-islanding (Centralized Island Monitoring Unit):**

- There will be a Centralized Island monitoring and control unit needs to be incorporated at SLDC Bihar for continuous monitoring of load generation balance in the island. It is necessary to maintain the load generation balance within the island for island stability.
- The control scheme will continuously monitor load generation imbalance and will trip identified feeders' priority wise if load generation imbalance goes beyond a certain limit and frequency reaches 48.7 Hz for 200 msec.

- **Islanding (2 stages):**

- When Frequency reaches 48.4 Hz, then with a delay of 500 msec, identified system will be islanded. For islanding, a number of tie lines need to be tripped to isolate the system from the grid. The command to trip the feeders will go from the Central master controller. As a back-up UFR relays may be installed in the identified feeders set at 48.4 Hz and 500 msec time delay.
- After islanding, another stage of feeder disconnection is also to be done if island frequency decreases. Three sub-stages are set after islanding and UFR relays will be installed on the identified feeders to get the desired load relief.
 - Stage 2A: 80 MW at 48.2 Hz
 - Stage 2B: 40 MW at 48.0 Hz
 - Stage 2C: 50 MW at 47.8 Hz

Feeders selected for pre-islanding disconnection will be identified as per below logic:



Deliberation in 53rd TCC meeting

SLDC Bihar apprised that DPR of the islanding scheme has been prepared with tentative cost implications of around 5.5-6 Cr.

TCC decision

- ✓ TCC recommended the Patna Islanding Scheme and referred to ERPC for concurrence.
- ✓ TCC advised Bihar to put up the proposal for funding from PSDF.

Deliberation in 53rd ERPC meeting

ERPC agreed with the proposal of Patna Islanding Scheme and advised Bihar to go ahead with the implementation scheme in a time bound manner.

SLDC Bihar may update the present status along with future action plan. Members may discuss.

Deliberation in the meeting

SLDC Bihar updated:

- ✓ A letter on estimated cost of the islanding scheme has been shared with the nodal agency for PSDF(NLDC).
- ✓ DPR will be shared with PSDF after rate justification from different agencies.

OCC Decision

OCC advised Bihar SLDC to expedite submission of Final DPR of Patna islanding scheme along with detailed cost breakup for PSDF grant.

2.13 Frequency Response Performance (FRP) of Generators: ERLDC

As per IEGC 2023 Clause 30, Sub clause 10(q):

NLDC, RLDCs and SLDCs shall grade the median Frequency Response Performance annually, considering at least 10 reportable events. **In case the median Frequency Response Performance is less than 0.75 as calculated as per mentioned table, NLDC, RLDCs, SLDCs, as the case may be, after analysing the FRP shall direct the concerned entities to take corrective action. All such cases shall be reported to the concerned RPC for its review.**

TABLE C: FREQUENCY RESPONSE CRITERIA

S. N	Performance*	Grading
i.	$FRP \geq 1$	Excellent
ii.	$0.85 \leq FRP < 1$	Good
iii.	$0.75 \leq FRP < 0.85$	Average
iv.	$0.5 \leq FRP < 0.75$	Below Average
v.	$FRP < 0.5$	Poor

**Provided that for wind/solar generating stations and state control areas with internal generation less than 100 MW or annual peak demand less than 1000 MW, the FRP grading shall be indicative only.*

ERLDC is preparing the annual FRP grading response for all ISGSs and IPPs and based on that corrective action plan will be intimated. SLDCs are requested to ensure calculation of annual frequency response grading of eligible units in their jurisdiction and submit the same with ERLDC and ERPC by 15th April 2025.

ERLDC may explain. Members may discuss.

Deliberation in the meeting

ERLDC mentioned that they are calculating median FRP of all the generating units in their jurisdiction. SLDC also needs to do the same for all the generation within their jurisdiction and share the same with ERLDC and ERPC. Based on the final Annual median FRP suitable action plan needs to be finalized.

OCC Decision

- OCC advised all SLDCs to share the FRP data for all the reportable events during 2024-25 to ERLDC by 15th April-2025.
- OCC advised all Generators to carry out Primary frequency response testing as a corrective action if FRP falls below the specified limits as per IEGC 2023.

2.14 Verification of Average Generation/Demand data for assessment of Frequency Response Obligation (FRO) for FY 2025-26: ERLDC

- IEGC Reg. 30(10) (f) mandates that the “Frequency Response Obligation (FRO) of each control area shall be assessed by NLDC and the same shall be informed to all control areas by 15th of March every year for the next financial year.” Accordingly, NLDC is in the process of calculating the FRO for each control area for the FY 2025-26. Annexure-2 of IEGC provides that the minimum FRO of each control area in MW/Hz shall be calculated as:

FRO = (Control Area average Demand + Control Area average Generation) * minimum all India Target Frequency Response Characteristic/ (Sum of average demand of all control areas + Sum of average generation of all control areas)

- Accordingly, ERLDC has computed the Average generation of all the ISGS/Central Sector IPPs and Average Generation and Demand data of State control areas of Eastern Region which has been communicated to the concerned utilities for verification. Till now the verification status is given below:

Control Area	Generation	Verification Status	Control Area	Generation	Verification Status
BIHAR	433		FARKKA-I & II	1119	
DVC	4655		FKSTPP-III	339	
JHARKHAND	471		GMR	552	
ODISHA	3284		JSW Utkal Energy	186	
SIKKIM	0		JITPL	930	
WEST BENGAL	5600		KAHALGAON -stg 1	615	
APNRL	398		KAHALGAON -stg 2	1145	
BARH-STG-1	766		MPL	809	
BARH-STG-2	1001		NKSTPP	970	
BRBCL	743		NPGC	1474	
DARLIPALLI	1296		TALCHER stg-1	700	
	Received				
	Not received				

All the ISGS/IPP and states are requested to verify the data at the earliest for finalization of FRO calculation for the year 2025-26 at the earliest.

ERLDC may explain. Concerned constituents may update. Members may discuss.

Deliberation in the meeting

ERLDC explained that for finalizing the primary frequency response obligation for 2025-26, control area wise average generation and demand data is required for the Calendar year of 2024. ERLDC mentioned that as per the data available at ERLDC, control area wise average demand and generation is already calculated and shared with each utility for verification. Following data are used for calculating the average value of Demand and generation:

1. Daily power supply position demand MU is collected for 365 days and then average MW demand is calculated for state control area
2. Daily power supply position state internal generation MU is collected for 365 days and then average MW generation(A) is calculated for state control area. Similarly for internal RE generation(B) of state is calculated. State internal average generation without RE is then calculated by A-B.
3. For all generating station under RLDC jurisdiction which are eligible for primary frequency response, average generation is calculated by 15min SEM data average over one year time period.

OCC Decision

OCC advised all SLDCs to submit to ERLDC the average generation and demand data i.r.o respective control areas for calendar year 2024 within next 10 days.

2.15 Generation target for FY 2025-26: ERPC

THERMAL

- Annual assessment and finalization of the Generation Programme and Planned Maintenance Schedules of generating units is undertaken by CEA every year. This process involves fixing up the Overall Generation Target for the country (involving Fuel-wise fixation of Generation Target also) based on last year generation, anticipated demand, likely economic growth etc.
- Following this, Fuel Wise target will be allocated to the various generating stations based on their past performances, planned maintenance schedule and the future planning as submitted by the respective generating station.

☐ In this regard, all power generating stations are requested to furnish the below mentioned details as per enclosed formats

a) Unit-wise monthly generation proposed during **2025-26** taking into account likely fuel availability, the anticipated loss of generation on account of various factors such as grid constraint, low schedule/ Reserve shut down due to high cost, coal/lignite quality etc., if any

b) The Unit-wise schedule of planned Maintenance for the year 2025-26.

On approval of planned maintenance schedule by the respective RPCs (Regional Power Committees), same shall be taken up by CEA(GM Division) to facilitate planning at All India level.

220th OCC Decision

- OCC advised all thermal generating units of ER to submit the details of unit-wise planned maintenance and anticipated generation(MU) for FY 2025-26 at the earliest.

223rd OCC Decision

- OCC informed all thermal GENCOs that submission of generation target of their respective units is crucial for finalizing Pan-India generation target and Resource adequacy planning by Ministry of Power.
- OCC advised all thermal generating units of ER to submit the details of unit-wise planned maintenance and anticipated generation(MU) for FY 2025-26 within a week positively.

All thermal GENCOs of ER who have not yet furnished the generation target are earnestly request to update the details.

Deliberation in the meeting

OCC Decision

OCC advised OPGC, TVNL, IndBharat, Vedanta, Jojobera TPS and Hiranmayee TPP to furnish generation target w.r.t their units for FY 2025-26 immediately.

2.16 Self-Audit in compliance with clause 56(2) of IEGC: ERLDC

- IEGC 2023 came into effect from 01st October 2023. As per IEGC Cl. 56(2) of the regulation, all users, CTU, STUs, NLDC, RLDCs, RPCs and SLDCs, power exchanges, QCA, SNAs shall conduct annual self-audits to review compliance of these regulations and submit the reports by **31st July** of every year. The self-audit reports by users, QCAs, SNAs shall be submitted to the concerned RLDC or SLDC, as the case may be.
- All users of ERLDC are requested to conduct self-audit as mandated by IEGC and submit the report to ERLDC by **31st July 2025** for FY25.

Members may discuss.

Deliberation in the meeting

OCC Decision

OCC advised all concerned utilities and organisations to submit annual self-audit report to ERLDC in compliance to IEGC 2023 regulations by 31st July 2025.

2.17 Shutdown request by DANS Energy

As per **deliberation in the 221st OCC meeting**:

DANS energy informed that due to prevailing winter season, water availability is not sufficient for full generation of their hydro units.

OCC Decision

OCC approved the shutdown requests of the hydro generating units of Tashiding(unit-01)(i.e 01-12-2024 to 14-01-2025) and Jorethang(unit-02) HEPs(02-12-2024 to 15-01-2024).

Further 222nd OCC Decision

OCC granted consent to the proposed shutdown of units of Jorethang Loop HEP from 1st February 2025 to 15th February 2025.

However, as per the directive from the Office of the District Collector, Soreng, Government of Sikkim (attached letter), we were unable to proceed with the shutdown as scheduled. The letter states that the ongoing river training/protective work at the confluence of River Rangeet and River Rammam would be impacted if water was released during this period. Consequently, DANS Energy is required to defer the shutdown beyond 17.02.2025.

Revised shutdown requests are as follows:

Tashiding HEP - Unit 1 & Unit-2

Planned Shutdown Dates: 24-03-2025 to 28-03-2025 (5 days)

Reason: Annual maintenance, major overhauling, and repair activities.

Jorethang Loop HEP

Planned Shutdown Dates: Both Unit 1 & Unit-2 tentatively scheduled between 5th March to 15th March 2025 (subject to approval from local authorities). The shutdown will be for a period of 15 days.

Reason: Annual maintenance and major overhauling activities.

These shutdowns are essential to ensure the reliability and operational efficiency of the units..

DANS Energy may update. Members may discuss.

Deliberation in the meeting

OCC Decision

The proposed shutdowns of Tashiding HEP(both units) & Jorethang Loop HEP(both units) were consented to if availed within March 2025. DANS Energy was advised to finalize the exact dates for availing the planned shutdowns in March 2025.

2.18 Periodic Testing of power system elements: ERPC

As mandated in **IEGC 2023, 40.1 & 40.2**, periodic tests shall be carried out on power system elements to ascertain the correctness of mathematical models used for simulation studies as well as ensuring desired performance during an event in the system.

Relevant portion of clause is as below:

Quote:

“

40. PERIODIC TESTING

.....

40.2 (a) The owner of the power system element shall be responsible for carrying out tests as specified in these regulations and for submitting reports to NLDC, RLDCs, CEA and CTU for all elements and to STUs and SLDCs for intra-State elements.”

40.2 (b) “All equipment owners shall submit a testing plan for the next year to the concerned RPC by 31st October to ensure proper coordination during testing as per the schedule. In case of any change in the schedule, the owners shall inform the concerned RPC in advance.

.....”

Unquote

In 217th OCC Meeting held on 24.07.2024, the matter was discussed in detail and OCC advised all the generators & owners of HVDC/FACTS devices to strictly adhere to the IEGC 2023 guidelines & submit the required testing data & plan to ERPC at the earliest (as per clause 40.2.(b).

Power Elements	System	Tests	Applicability
Synchronous Generator		(1) Real and Reactive Power Capability assessment. (2) Assessment of Reactive Power Control Capability as per CEA Technical Standards for Connectivity (3) Model Validation and verification test for the complete Generator and Excitation System model including PSS. (4) Model Validation and verification of Turbine/Governor and Load Control or Active Power/ Frequency Control Functions. (5) Testing of Governor performance and Automatic Generation Control.	Individual Unit of rating 100MW and above for Coal/lignite, 50MW and above gas turbine and 25 MW and above for Hydro.
HVDC/FACTS Devices		(1) Reactive Power Controller (RPC) Capability for HVDC/FACTS (2) Filter bank adequacy assessment based on present grid condition, in consultation with NLDC. (3) Validation of response by FACTS devices as per settings.	To all ISTS HVDC as well as Intra-State HVDC/FACTS, as applicable

223rd OCC Decision

All GENCOs of ER who have not yet submitted details were advised to furnish the testing plan i.r.o individual synchronous generators in the shared google sheet within a week positively in compliance to IEGC 2023.

- ERLDC was requested to maintain a consolidated database on testing schedule of all generating utilities of ER.

Details still awaited from OPGC, OHPC, Ind Bharat, Vedanta, HEL, Hiranmoyee Energy. CESC submitted incomplete details.

All these Gencos are requested to submit schedule in the Google sheet.
<https://docs.google.com/spreadsheets/d/1m6KCKONdObMhre9-1me1kvHTEBYUdXUOISYdn5FR4fM/edit?gid=0#gid=0>

All GENCOs and HVDC/FACTS owners may update.

Deliberation in the meeting

OCC Decision

OCC urged OPGC, OHPC, Ind Bharat, Vedanta, HEL and Hiranmoyee Energy to update the testing plan of their generators in the google sheet:
<https://docs.google.com/spreadsheets/d/1m6KCKONdObMhre91me1kvHTEBYUdXUOISYd5FR4fM/edit?gid=0#gid=0>. CESC was advised to update the complete details of testing plan.
 This is in compliance to the provisions of IEGC 2023

3. PART-C: ITEMS FOR UPDATE/FOLLOW-UP/INFORMATION

3.1. ER Grid performance during January 2025.

The average and maximum consumption of Eastern Region and Max/Min Demand (MW), Energy Export for the month January-2025 were as follows:

AVERAGE CONSUMPTION (MU)	MAXIMUM CONSUMPTION(MU)/ DATE	MAXIMUM DEMAND (MW)	MINIMUM DEMAND (MW)	SCHEDULE EXPORT	ACTUAL EXPORT
		DATE / TIME	DATE / TIME	(MU)	(MU)
465.1 MU	481.8 MU, 30.01.2025	24616 MW, 21.01.2025 at 18:01 Hrs.	15142 MW, 02.01.2025 at 03:47 Hrs.	4625	4588

ERLDC/ERPC may highlight the performance of the ER grid.

Deliberation in the meeting

The grid performance of ER for the month of January 2025 was highlighted.

3.2. Update on Reconductoring of ISTS lines under Eastern Region Expansion Scheme-44: ERPC

- ❖ Several 220 kV transmission lines and substations were implemented in Indian grid along with cross border lines for importing power from Chukha Hydro Electric Plant in Bhutan. The generating station was commissioned in years 1986-88 and the transmission system is now more than 35 years old. Considering the age of conductors and increase in conductor snapping incidences, reconductoring of these transmission lines has become necessary.
- ❖ The matter was deliberated in various OCC forums as well as in 52nd TCC meeting of ERPC.
- ❖ In a meeting was convened by CEA under the chairpersonship of Member (Power System) on 27-08-2024, it was decided that matter of reconductoring of cross border lines will be separately taken up with Bhutan.
- ❖ However, reconductoring of ISTS portion of 220 kV corridor viz. Alipurduar (POWERGRID) – Falakata (WBSETCL) – Birpara (POWERGRID) – Binaguri (POWERGRID) – Siliguri (POWERGRID) – Kishanganj (POWERGRID) – Dalkhola (POWERGRID) – Gazole (WBSETCL) – Malda (POWERGRID), may be taken up under ISTS. Further, reconductoring of intra-state LILO portion of Birpara (POWERGRID) – Alipurduar (POWERGRID) 220 kV D/c line at Falakata (WBSETCL) and Dalkhola – Malda 220 kV D/c line at Gazol (WBSETCL) shall be carried out by WBSETCL matching with HTLS conductor of the main ISTS line in the matching timeframe.

Name of the scheme	Implementation timeframe	Implementation mode	Implementing agency	Estimated Cost (Rs. in Cr)
ERES-44	18 months (15 months on best effort basis) from the date of allocation	RTM	Powergrid	385.77

WBSETCL works associated with reconductoring of ISTS lines

- ❖ In the NCT(National Committee on Transmission) meeting dated 23.10.2024 , the following were decided:
- ✦ WBSETCL shall reconductor their following lines sections under intra-state scheme matching with completion of ISTS scheme namely ERES-44:
- ✓ About 4 km intra-state portion of Alipurduar (POWERGRID) Falakata (WBSETCL) 220 kV D/C line at Falakata end with HTLS conductor of ampacity 1250 A along with necessary upgradation of associated 220 kV bay equipment at Falakata (WBSETCL) end commensurate with rating of HTLS (1250 A).
- ✓ About 4 km intra-state portion of Birpara (POWERGRID) - Falakata (WBSETCL) 220 kV D/C line at Falakata end with HTLS conductor of ampacity 1250 A along with necessary upgradation of associated 220 kV bay equipment at Falakata (WBSETCL) end commensurate with rating of HTLS (1250 A).
- ✓ About 2 km intra-state portion of Dalkhola (POWERGRID) - Gazole (WBSETCL) 220 kV D/C line at Gazole end with HTLS conductor of ampacity 1250 A along with necessary upgradation of associated 220 kV bay equipment at Gazole (WBSETCL) end commensurate with rating of HTLS (1250 A).
- ✓ About 2km intra-state portion of Gazole (WBSETCL) - Malda (POWERGRID) 220 kV D/C line at Gazole end with HTLS conductor of ampacity 1250 A along with necessary upgradation of associated 220 kV bay equipment at Gazole (WBSETCL) end commensurate with rating of HTLS (1250 A).
- ✦ WBSETCL will LILO the Dhalkola - Gazole 220 kV D/C line with 1250 A HTLS under their intra-state scheme for establishment of 220 kV level at their existing 132/33kV Raiganj (WBSETCL) S/S.
- ✦ ISTS licensee and WBSETCL shall coordinate for reconductoring of their respective portion of the lines matching with completion schedule of this scheme.
- It is kindly requested that WBSETCL may note the scope of works (as provided in the minutes of NCT) and coordinate with POWERGRID for matching implementation of their works.
- The **progress report** may be shared on **monthly** basis to **CEA, ERPC** and **CTU**.

As per **223rd OCC** Deliberation

Powergrid ER-II updated:

A meeting was convened on 17.01.2025 with Bhutan to finalize the modalities of reconductoring of CTS in Bhutan portion. MOM of the same will be circulated.

223rd OCC Decision:

- OCC advised Powergrid ER-II to expedite the reconductoring works of CTS as per timeline approved in 24th NCT meeting and share monthly progress report of the same with CEA, CTU and ERPC.
- WBSETCL was advised to coordinate with Powergrid ER-II for carrying out reconductoring works on intra-state portions of 220 kV lines under CTS matching with the timeframe of ISTS lines. This is in compliance to decisions of 24th NCT meeting. Both WBSETCL and Powergrid need to finalize modalities bilaterally and update in next OCC.

Update as per **53rd TCC**:

Powergrid ER-II submitted :

- MOU needs to be signed with WNSETCL i.r.o of cost sharing modalities for reconductoring of the intra-state portions of 220 kV lines.
- Presently tendering is under progress and reconductoring of all approved 220 kV lines to be completed tentatively by May 2026.

WBSETCL and Powergrid may respond. Members may discuss.

Deliberation in the meeting

Powergrid apprised:

- ✓ *NIT for reconductoring under ERES-44 scheme shall be floated on 04.3.2025.*
- ✓ *Modalities of reconductoring in Bhutan portion could not be finalized yet due absence of response from Bhutan end after bilateral meeting being held.*
- ✓ *MOU has already been done with WBSETCL for reconductoring in intra-state portions of West Bengal network.*

OCC Decision

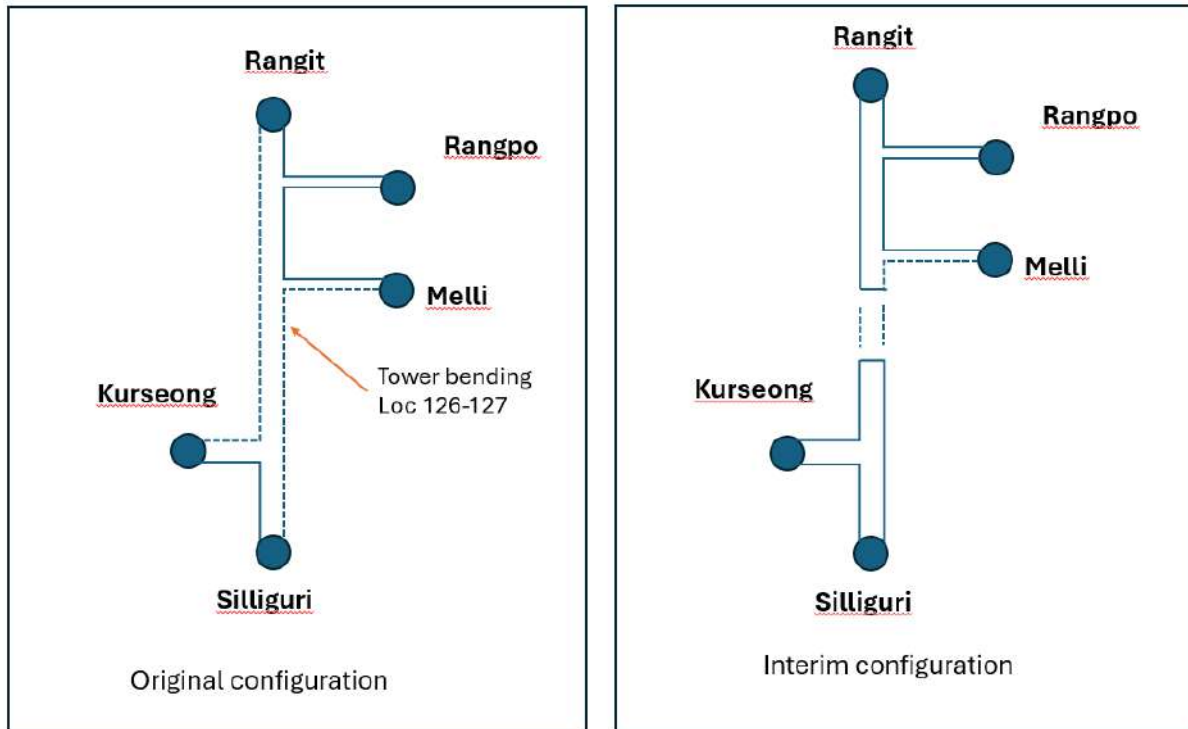
- *Powergrid was urged to expedite the tendering process of reconductoring works under ERES-44.*
- *Powergrid was advised to write a letter to Bhutan Power system operator with copy to CEA and MOP for expediting finalization of modalities of reconductoring with Bhutan.*

3.1. Update on Restoration of 132kV Rangit-Kurseong & 132kV Siliguri-Melli-Rangpo lines: ERLDC

- Due to incessant rain and several landslides, towers at loc. 125-128 of **132 kV Rangit-Kurseong** and **132 kV Siliguri-Melli** got badly affected. Out of which tower at loc. 126,127 got severely damaged. Both the lines were switched on 5th October 2024 on request of PowerGrid.
- Consequently, **Kurseong** and **Melli** (Kalimpong source) are fed through single source of **Siliguri** and **Rangpo** respectively. To ensure reliable power supply at Melli & Kurseong, ERLDC conducted one meeting on **08.10.2024** (online mode) with participants from ERPC, ERLDC, West Bengal SLDC, Sikkim, Powergrid and NHPC Rangit.
- Considering the difficulties & time requirements due to hilly terrain for restoration of the said portion, temporary reconfiguration of these lines was explored to extend additional sources to Melli & Kurseong. It was decided that part of the healthy line of **132 kV Siliguri-Melli** will be reconfigured as **132 kV Siliguri-Kurseong ckt2** as a second source of

Kurseong and another healthy portion of **132 kV Siliguri-Melli** will be reconfigured as **132 kV Rangit-Melli** for a second source of Melli.

- After necessary reconfiguration, **132 KV Siliguri-Kurseong-II (interim)** arrangement charged on **9th October** and **132kV-Rangit-Melli (interim)** has been charged tentatively on **22nd October**. POWERGRID intimated that it would take 15-20 Days to restore the original configuration after rectifying damaged towers.



Update as per **53rd TCC**:

Powergrid ER-II apprised:

- ✓ LOA already placed for new foundation and associated works.
- ✓ New area for foundation has already been identified and excavation will start. subject to clearance from Soom Tea Garden (Owner of the land).
- ✓ As per meeting with GM/Soom Tea Garden (06.02.2025), clearance will be available after concurrence from Board members, tentatively by 13.02.2025.
- ✓ Required Tower parts (400KV)/(DD+18) are already diverted (From Maithon) and available at Binaguri S/S.
- ✓ Letter is given to WBSETCL for lowering OPGW on subject span and subsequent discussion already done. Will be done by WBSETCL once work gets started at said location.

Completion Target: End of April, 2025

Powergrid and WBSETCL may update. Members may discuss.

Deliberation in the meeting

Powergrid ER-II updated:

- ✓ Persistent ROW issues have been resolved to a larger extent in the area of Soom Tea Garden.
- ✓ Confirmation from District collector is pending for commencing works in that area. A meeting has been scheduled in this regard District collector with on 26.02.2025.

OCC Decision

OCC noted the proceedings and urged Powergrid to expedite for adhering to the completion target i.e End of April, 2025.

3.2. Regarding Non-Submission of Forecasting Data from States: ERLDC

The **Clause 2 of Regulation 31 of IEGC 2023** has mandated all the SLDCs to timely submit the demand estimate data to the respective RLDC and RPC.

The demand estimation data provided by SLDCs will be required in resource adequacy planning and regional load forecasts conducted by the RLDC.

Currently, the day ahead data is regularly received from all the states except Sikkim.

223rd OCC decision:

- OCC advised all SLDCs for strictly adhering to the schedule of demand estimation as mandated in IEGC 2023, timely sharing with ERLDC in specified format as well as uploading of forecasting error on their respective websites.
- SLDCs who are submitting day ahead forecast were advised to also share the forecasting data for their respective control areas on weekly as well as monthly basis with ERLDC.
- All SLDCs were urged to regularly furnish resource adequacy data besides demand forecast.
- SLDC Odisha was advised to expedite implementation of the demand forecasting software and positively update the status in next OCC.

Latest Forecast receipt status is shown below:

	Status of Furnishing of Day Ahead Forecast data by ER States																															
Bihar																																
Jharkhand																																
DVC																																
Odisha																																
West Bengal																																
Sikkim																																
																													</			

	Status of Furnishing of Day Ahead Resource adequacy data by ER States																														
Bihar																															
Jharkhand																															
DVC																															
Odisha																															
West Bengal																															
Sikkim																															
	01-01-25	02-01-25	03-01-25	04-01-25	05-01-25	06-01-25	07-01-25	08-01-25	09-01-25	10-01-25	11-01-25	12-01-25	13-01-25	14-01-25	15-01-25	16-01-25	17-01-25	18-01-25	19-01-25	20-01-25	21-01-25	22-01-25	23-01-25	24-01-25	25-01-25	26-01-25	27-01-25	28-01-25	29-01-25	30-01-25	31-01-25

Status of Furnishing of Week Ahead Demand Forecast data by ER States					
Bihar					
Jharkhand					
DVC					
Odisha					
West Bengal					
Sikkim					
	30.12.24 to 05.01.25	06.01.25-12.01.25	13.01.25-19.01.25	20.01.25-26.01.25	27.01.25-02.02.25

Status of Furnishing of Week Ahead Resource Adequacy data by ER States					
Bihar					
Jharkhand					
DVC					
Odisha					
West Bengal					
Sikkim					
	30.12.24 to 05.01.25	06.01.25-12.01.25	13.01.25-19.01.25	20.01.25-26.01.25	27.01.25-02.02.25

Status of Furnishing of Month Ahead Forecast data by ER States	
Bihar	
Jharkhand	
DVC	
Odisha	
West Bengal	
Sikkim	
	January

Status of Furnishing of Month Ahead Resource Adequacy data by ER States	
Bihar	
Jharkhand	
DVC	
Odisha	
West Bengal	
Sikkim	
	January

ERLDC may explain and all SLDCs may update. Members may discuss.

Deliberation in the meeting

OCC Decision

- OCC advised all SLDCs for strictly adhering to the schedule of demand estimation as mandated in IEGC 2023, timely sharing with ERLDC in specified format as well as uploading of forecasting error on their respective websites.
- SLDCs who are submitting day ahead forecast were advised to also share the forecasting data for their respective control areas on weekly as well as monthly basis with ERLDC.
- All SLDCs were urged to regularly furnish resource adequacy data besides demand forecast.

3.3. Non-Submission of FRC data in stipulated timeframe: ERLDC

Adhering to IEGC clauses **30.8** and **30.10.(a)** to **30.10.(q)**, generating stations within the Eastern region are required to submit essential data to ERLDC within two days of receiving a notification regarding a reportable frequency event. Additionally, according to clause 30.10.(n), all control areas within the eastern region must assess their frequency response characteristics and share the evaluation, along with high-resolution data, with the ERLDC. Therefore, timely submission of primary response data is crucial for compliance with the IEGC.

223rd OCC decision:

All generators were advised to regularly share high resolution data against each reportable frequency event with ERLDC on time to facilitate accurate assessment of FRP for respective control areas.

The latest data receipt status is given below: (as on **19.02.2025**):

STATIONS	DATE	04.01.2025
	TIME	19:23
FSTPP #STG 1 & 2	ISGS	
FSTPP # STG 3	ISGS	
KhSTPP #STG 1	ISGS	
KhSTPP #STG 2	ISGS	
TSTPP #STG 1	ISGS	
Barh stage-1	ISGS	
Barh stage-2	ISGS	
BRBCL	ISGS	
Darlipalli	ISGS	
North Karanpura	ISGS	
NPGC	ISGS	
TEESTA V	ISGS	
GMR	CPP	
MPL	CPP	
ADHUNIK	CPP	

JITPL	CPP	
INDBHARAT	CPP	
TEESTA III	CPP	
Bihar	STATE	
Jharkhand	STATE	
DVC	STATE	
OPTCL	STATE	
WB	STATE	
Updated as on	19.02.2025	
	Received	
	Not Received	
	Plant Out	
	Data freeze at plant	

Hence all are again requested to follow the stipulated timeline and submit the data to ERLDC and also fill the google sheet below to include the email address where notifications of reportable events should be sent.

https://docs.google.com/spreadsheets/d/1slvAOmQIEQVIMn0LnB78eKMa2sz2QYICZ-sPEpeV_jk/edit?usp=sharing

ERLDC may explain. Members may discuss.

Deliberation in the meeting

- ✓ *All generators were advised to regularly share high resolution data against each reportable frequency event with ERLDC on time to facilitate accurate assessment of FRP for respective control areas.*
- ✓ *All generating utilities were also urged to update the google sheet (link mentioned above) with email address where notifications of reportable events will be shared.*

3.4. Mock Black Start: ERLDC

- As per IEGC Reg. 34.3: A mock trial run of the procedure for different sub-systems including black-start of generating units along with grid forming capability of inverter-based generating station and VSC-based HVDC black-start support at least once a year under intimation to the concerned SLDC and RLDC.
- Eastern region has 16 hydro power plants, which has capability to play a crucial role during restoration after any grid disturbance. Mock black start testing along with grid forming capability is being carried out on yearly basis, as mandated by IEGC reg 34.3, to ensure the capability & readiness of those generators for any contingency.
- Also, diesel generator sets and other standalone auxiliary supply source to be used for black start shall be tested on a weekly basis and the test reports are to be shared to the concerned SLDC, RLDC and NLDC on a quarterly basis.

- As per IEGC Reg. 34.4: Simulation studies are to be carried out by each user in coordination with RLDC for preparing, reviewing and updating the restoration procedures considering the following:

- (a) Black start capability of the generator;
- (b) Ability of black start generator to build cranking path and sustain island;
- (c) Impact of block load switching in or out;
- (d) Line/transformer charging;
- (e) Reduced fault levels;
- (f) Protection settings under restoration condition

So far, Balimela, Burla, U. Indravati, TLDP-IV, and Subarnarekha have completed their mock black start tests, while Jorethang and Tashiding have confirmed tentative dates for FY25. The remaining generators are yet to schedule their tests and are requested to confirm their mock drill dates. Status of mock black start is as follows:

Sl. No.	Name of Hydro Station	2024-25 Actual Date of Test	Tentative date as on 18.02.25
1	U. Kolab		Yet to be informed
2	Balimela	15 th January 2025	
3	Rengali		Yet to be informed
4	Burla	December-24	
5	U. Indravati	Sep-24	
6	Maithon	December-24	
7	TLDP-III		Yet to be informed
8	TLDP-IV	December-24	
9	Subarnarekha	3 rd December 2024	
10	Teesta-V	N/A	N/A
11	Chuzachen		Yet to be informed
12	Teesta-III	N/A	N/A
13	Jorethang		25 th February 2025
14	Tashiding		29-31 March 2025
15	Dikchu	N/A	Yet to be informed
16	Rongnichu		Yet to be informed

The rest of the generators are requested to confirm dates for black start of each generating unit. Also, the users are requested to share the data required simulation studies before the scheduled date of mock drill.

Members may note.

Deliberation in the meeting

OCC Decision

- *OCC advised all black start capable hydro generating units of ER to update their schedule of mock black start to ERLDC at the earliest. This is in compliance to IEGC 2023 (CERC).*
- *OCC further opined that in case of non-receipt of further update by respective hydro generating units the proposed tentative schedule of mock black start may be considered as final. Thereafter all black start capable hydro units shall have to conduct mock black start at least once in a year as mandated in IEGC 2023.*

3.5. Commissioning Status of ADMS: ERLDC

The automatic demand management scheme (ADMS) has been already commissioned in West Bengal, DVC, Odisha, and Jharkhand and partially implemented by Bihar. In the 216th OCC meeting the forum advised Bihar to share detailed action plan for implementation of additional 400 MW load under ADMS. DVC has shared revised list of feeders under ADMS after exclusion of feeders present in CTPS islanding scheme. BSPTCL yet to update the status.

Bihar may update. Members may discuss.

Deliberation in the meeting

OCC Decision

OCC advised BSPTCL to share the feeder list of ADMS with ERLDC positively within a week.

4. PART-D: OPERATIONAL PLANNING

4.1. Anticipated power supply position during March-2025

The abstract of peak demand (MW) vis-à-vis availability and energy requirement vis-à-vis availability (MU) for the month of March 2025 is prepared by ERPC Secretariat (**Annexure D.1**) on the basis of LGBR for 2024-25 and feedback of constituents, keeping in view that the units are available for generation and expected load growth etc.

Members may update.

Deliberation in the meeting

*The updated anticipated power supply position for March 2025 is provided at **Annexure D.1**.*

4.2. Major Thermal Generating Units/Transmission Element outages/shutdown in ER Grid (as on as on 16-02-2025)

a) **Thermal Generating Stations outage report:**

SL No	STATION	STATE	AGENCY	UNIT NO	CAP ACIT Y (MW)	REASON(S)	OUTAGE DATE
1	BARAUNI TPS	BIHAR	NTPC	7	110	Initially unit was tripped due to poor condenser vacuum. After that Operation permanently discontinued w.e.f. 31.03.2024	19-Jul-2023
2	BARAUNI TPS	BIHAR	NTPC	6	110	Initially unit was tripped due to Low vacuum. After that operation permanently discontinued w.e.f. 31.03.2024	22-Jul-2023
3	Sterlite	ODISHA	SEL	2	600	Initially reported APH problem. Later on dated 11.02.25 mail received, Turbine box up under progress and Boiler R&M being done.	15-Dec-2024
4	IB.TPS	ODISHA	OPGC	2	210	Annual Overhauling	07-Jan-2025
5	SAGARDIG HI	WEST BENGAL	WBPDC	4	500	Annual Overhauling	10-Jan-2025
6	MEJIA TPS	DVC	DVC	3	210	Capital Overhauling	10-Feb-2025
7	KOLAGHAT	WEST BENGAL	WBPDC	6	210	Generator seal liner temp High	16-Feb-2025
8	NABINAGAR(NPGC)	BIHAR	NTPC	3	660	Separator pressure high	17-Feb-2025
9	TENUGHAT	JHARKHAND	TVNL	2	210	High turbine vibration.	18-Feb-2025
10	MEJIA TPS	DVC	DVC	7	500	Boiler Tube Leakage	18-Feb-2025

All Generating stations are requested to update expected restoration time and reason outage to ERLDC/ERPC on weekly basis in case of any change at their end.

b) Major Generating stations Out on Reserve Shutdown due to low system demand:

SL No	STATION	STATE	AGENCY	UNIT NO	CAPACITY (MW)	REASON(S)	OUTAGE DATE
1	SOUTHERN	WEST BENGAL	CESC	1	67.5	Low system demand	14-Dec-2024
2	SOUTHERN	WEST BENGAL	CESC	2	67.5	Low system demand	11-Dec -2024

c) Hydro Unit Outage Report:

S. NO	STATION	STATE	AGENCY	UNIT NO	CAPACITY (MW)	REASON(S)	OUTAGE DATE
1	TEESTA STG III Hep	SIKKIM	TUL	1	200	Sudden cloudburst at glacier fed LOHNAK Lake followed by huge inrush of water in Teesta River and damage of Teesta III Dam & downstream Powerhouses	04-Oct-2023
2	TEESTA STG III Hep	SIKKIM	TUL	2	200		
3	TEESTA STG III Hep	SIKKIM	TUL	3	200		
4	TEESTA STG III Hep	SIKKIM	TUL	4	200		
5	TEESTA STG III Hep	SIKKIM	TUL	5	200		
6	TEESTA STG III Hep	SIKKIM	TUL	6	200		
7	DIKCHU Hep	SIKKIM	SKPPL	2	48	Sudden cloudburst at glacier fed LOHNAK Lake followed by huge inrush of water in Teesta River and damage of Teesta III Dam & downstream Powerhouses	04-Oct-2023
8	TEESTA HPS	SIKKIM	NHPC	1	170	Sudden cloudburst at glacier fed LOHNAK Lake followed by huge inrush of water in Teesta River and damage of Teesta III Dam & downstream Powerhouses	04-Oct-2023
9	TEESTA HPS	SIKKIM	NHPC	2	170		
10	TEESTA HPS	SIKKIM	NHPC	3	170		
11	RANGIT HPS	SIKKIM	NHPC	3	20	Initially unit was taken out for replacement of Main Inlet Valve (MIV) Seal, later unit was taken under Capital Overhauling from 00:00 hrs of 27.12.2024.	25-Dec-2024
12	CHIJLIMA HPS / HIRAKUD II	ODISHA	OHPC	1	24	Capital Overhauling	15-Dec-2023
13	INDRAVATI	ODISHA	OHPC	3	150	For replacement of Main Inlet Valve (MIV)	17-Dec-2024

14	BALIMELA HPS	ODISHA	OHPC	6	60	Initially unit was out due to Severe water leakage from turbine, later unit was taken under Repair and maintenance work from 00:00 hrs of 16.01.25	06-Jan-2025
15	U. KOLAB	ODISHA	OHPC	1	80	Capital Overhauling	07-Jan-2025
16	BALIMELA HPS	ODISHA	OHPC	5	60	Repair and maintenance work	16-Jan-2025
17	DIKCHU Hep	SIKKIM	SKPPL	1	48	Installation of new 132 kV GIS bay work.	20-Jan-2025
18	CHUZACHEN	SIKKIM	GATI	2	55	Annual Overhauling	06-Feb-2025
19	RENGALI HPS	ODISHA	OHPC	4	50	Annual Maintenance	10-Feb-2025
20	RONGNICHU	SIKKIM	MBPGCL	1	56.5	Annual maintenance	14-Feb-2025
21	RONGNICHU	SIKKIM	MBPGCL	2	56.5	Annual maintenance	14-Feb-2025

d)Long outage report of transmission lines (As on 16.01.2025):

Transmission Element / ICT	Outage From	Reasons for Outage
220/132KV 100 MVA ICT II AT LALMATIA	22.01.2019	220/132KV, 100MVA Transformer (NTPC side) is charged on 07.02.2024 from HV side on no load. Now, it is in idle charged condition
220KV-FSTPP-LALMATIA-I	21.04.2021	Two nos. of tower collapsed on 29.05.2024 near to Lalmatia GSS in the Loc. No. 246 & 247. Presently 220 kV Farakka-Lalmatia line is charged (from loc no 241 to loc 84) at 132 kV voltage level for anti-theft purpose by tapping at loc. No. 100-101.
220KV-WARIA-BIDHANNAGAR-1 & 2	08.06.2022	To control overloading of 220 kV Waria-DSTPS (Andal) D/C line
132KV-BARHI-RAJGIR-1	25.03.2023	Dismantling of tower no. 227, 228, and 229 crossing the premises of Mahabodhi Cultural centre along with Destraining of conductor of both circuits and Earth wire between tension tower no. 218-237 in same line.
132KV-NALANDA-BARHI(DVC)-1	25.03.2023	
400KV-RANGPO-TEESTA-V-1 & 2	04.10.2023	Tower near gantry of Teesta V powerhouse collapsed due to sudden cloudburst at glacier fed LOHNAK Lake followed by huge inrush of water in TEESTA river and damage of Teesta III Dam & downstream Powerhouses
400KV-TEESTA-III-RANGPO-1	04.10.2023	Hand tripped from Teesta-III end due to sudden cloudburst at glacier fed LOHNAK Lake followed by huge inrush of water in TEESTA river and damage of Teesta III Dam & downstream Powerhouses
400KV-TEESTA-III-DIKCHU-1	04.10.2023	
400KV-JHARSUGUDA-ROURKELA-4	01.04.2024	Reconductoring work

132KV-RANGPO-SAMARDONG-1	22-05-2024	Rangpo: Y-N fault with fault distance 0.157 km 14.562kA Samardong: NA
220KV-RAJARHAT-NEW TOWN(AA-II)- 2	10-07-2024	Initially line out due to rectification of gas leakage problem from B-Ph breaker pole. Line declared under breakdown after charging attempt after return of shutdown. After that fault found in b-phase cable.
132KV-RANGPO-SAMARDONG-2	02-08-2024	132/66/11kV Samardong ss have become inaccessible due to continuous raining and landslides. It is very difficult for round the clock deployment of shift manpower due to road non-accessibility
400KV/220KV 315 MVA ICT 2 AT INDRAVATI.	09-09-2024	Tripped due to Over Flux protection operated
400KV/220KV 315 MVA ICT 1 AT NORTH KARANPURA	12-09-2024	Tripped on Differential protection
132KV-MADHEPURA (BH)- SAHARSA(PMTL)-1	23.09.2024	To control loading on 132kV Madhepura-Saharsa line
132KV-MELLI-SILIGURI-1	05-10-2024	S/d for inspection of tower of Loc.127 found twisted due to heavy landslide & heavy continuous rainfall in Soom Tea Garden under Darjeeling section. Line charged as 132 KV Siliguri-Melli II (Interim arrangement) at 19:20 hrs on 09-10-2024. This interim arrangement is obtained by horizontal jumpering at Loc-129 after disconnecting main jumper for both Rangit & Melli side.
132KV-RANGIT-KURSEONG-1	05-10-2024	S/d for inspection of tower of Loc.127 found twisted due to heavy landslide & heavy continuous rainfall in Soom Tea Garden under Darjeeling section. Line charged as 132 KV Siliguri-Melli II (Interim arrangement) at 19:20 hrs on 09-10-2024. This interim arrangement is obtained by horizontal jumpering at Loc-129 after disconnecting main jumper for both Rangit & Melli side.
400KV/220KV 315 MVA ICT 1 AT TSTPP	01-11-2024	Tripped on PRD protection
132KV-PATRATU-PATRATU-1 & 2	16-11-2024	Diversion/Heightening of line due to inadequate clearance from under construction railway Line by PVUNL
132KV-CHUZACHEN-RANGPO-1	29-11-2024	Rangpo : B-N ,Z-1, 7.8 KA, 5.61 KM
400KV-ALIPURDUAR (PG)- PUNASANGCHUN-JIGMELING-2	02-12-2024	SD Availed by Bhutan for rectify/Replace the LA for 400kV Jigmeling _Puna_ALI-1.
400KV-KHSTPP-BARH-2	07-12-2024	Uprating of bay & line equipments
400KV-ALIPURDUAR (PG)- PUNASANGCHUN-JIGMELING-1	10-12-2024	Jumper connection and interconnection removal at Kamichu
220KV-DALKHOLA (PG)-GAZOLE-1	06-01-2025	To control loading of 220kV Malda-Gazole D/C
220KV-DALKHOLA (PG)-GAZOLE-2	07-01-2025	To control loading of 220kV Malda-Gazole D/C
400KV/220KV 315 MVA ICT 2 AT MEJIA-B	20-01-2025	Tripped during charging of ICT#1 bay with cable from 220 kv GIS side

400KV-BINAGURI-TALA-2	24-01-2025	Binaguri end: R-N, F dist 125.3 kM, F Current Ir-3.26kA
400KV-ROURKELA-JHARSUGUDA-3	07-02-2025	Reconductoring work on OCB till 28/02/2025
400KV-ROURKELA-JHARSUGUDA-4	01-02-2025	Reconductoring work on OCB till 28/02/2025
132KV-CHUZACHEN-RANGPO-1	04-02-2025	Maintenance Activities
400KV-RANCHI-ROURKELA-2	30-01-2025	Bay upgradation
132KV-NAGARUNTARI-NABINAGAR-1	07-02-2025	Re-sagging of conductors at various locations on OCB till 25/02/2025
132KV-RANGPO-GANGTOK-1	17-02-2025	Gangtok: Y-B Fault, F/D - 2.6 km;F/C-8.3 kA, Line is under Patrolling.
400KV/220KV 315 MVA ICT 3 AT TSL KALINGANAGAR	17-02-2025	Annual maintenance of ICT
400KV-NEW PURNEA-KISHANGANJ-1	18-02-2025	Facilitating Erection of New Tower on Pile foundation
400KV-KISHANGANJ-NEW PURNEA-2	18-02-2025	Facilitating Erection of New Tower on Pile foundation

Transmission licensees/ Utilities are requested to update expected restoration date & work progress regarding restoration regularly to ERLDC/ERPC on monthly basis by 5th of each month so that status of restoration can be reviewed in OCC. Utilities are also requested to update outage of any elements within their substation premises like isolator/breaker to ERLDC/ERPC regularly. (Reported as per Clause 5.2(e) of IEGC).

Deliberation in the meeting

Members noted.

4.3. Commissioning of new units and transmission elements in Eastern Grid in the month of January -2025.

The details of new units/transmission elements commissioned in the month of January 2025 based on the inputs received from beneficiaries:

NEW ELEMENTS COMMISSIONED DURING January, 2025

उत्पादन इकाइयाँ / GENERATING UNITS

क्र. Sl. No.	स्थान Location / Pooling Station	मालिक/यूनिट का नाम OWNER/UNIT NAME	यूनिट संख्या/स्रोत Unit No/Source	संकलित क्षमता (मेगावाट) Capacity added (MW)	कुल/स्थापित क्षमता (मेगावाट) Total/Installed Capacity (MW)	दिनांक DATE	टिप्पणी Remarks
1	Jharsuguda / JSWEUL-400 kV	JSW Energy (Utkal) Limited	2	350	700	15-01-2025	

आई.सी.टी./जी.टी./एस.टी / ICTs/ GTs / STs

क्र. Sl. No.	एजेंसी/मालिक Agency/Owner	उप-केन्द्र SUB-STATION	आईसीटी संख्या ICT NO	वोल्टेज (केवी) Voltage Level (kV)	क्षमता (एमवीए) CAPACITY (MVA)	दिनांक DATE	टिप्पणी Remarks
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1	WBPDCL	SAGARDIGHI	ST- 5	400/11 kV	130	15-01-2025	
प्रेषण लाइन / TRANSMISSION LINES							
क्र. Sl. No.	एजेंसी/मालिक Agency/Owner	लाइन का नाम LINE NAME	लंबाई (किमी) Length (KM)	कंडक्टर प्रकार Conductor Type	दिनांक DATE	टिप्पणी Remarks	
1	LILO Portion - 10.5 Km (BSPTCL)	400KV-BARH-BAKHTIYARPUR(BH)-1	51.04	QUAD MOOSE	31-01-2025	first time charged	
2	LILO Portion - 10.5 Km (BSPTCL)	400KV-PATNA-BAKHTIYARPUR(BH)-1	62.204	QUAD MOOSE	31-01-2025	first time charged	
3	OCPL	33KV-DARLIPALLI (NTPC)-MANOHARPUR-2	12.20 (11.7 km OH line +0.5 km U/G XLPE Cable)	AAAC	06-01-2025	Dedicated for Manoharpur mines (OCPL)	
लिलो / प्रेषण लाइन की पुनर्व्यवस्था / LILO/RE-ARRANGEMENT OF TRANSMISSION LINES							
क्र. Sl. No.	एजेंसी/मालिक Agency/Owner	लाइन का नाम / लिलो पर Line Name/LILO at	लंबाई (किमी) Length (KM)	कंडक्टर प्रकार Conductor Type	दिनांक DATE	टिप्पणी Remarks	
1	WBSETCL	400KV-ARAMBAGH-PPSP-2	209	ACSR MOOSE	06-01-2025	First time charged after Reconfiguration of 400KV-PPSP-NEW PPSP-2 and 400KV-NEW PPSP-ARAMBAGH-2 as 400KV-PPSP-ARAMBAGH	
बस/लाइन रिएक्टर / BUS/LINE REACTOR							
क्र. Sl. No.	एजेंसी/मालिक Agency/Owner	एलेमेंट का नाम Element Name	उप-केन्द्र SUB-STATION	वोल्टेज (केवी) Voltage Level (kV)	दिनांक DATE	टिप्पणी Remarks	
NIL							
□ □ / BUS							
क्र. Sl. No.	एजेंसी/मालिक Agency/Owner	एलेमेंट का नाम Element Name	उप-केन्द्र SUB-STATION	वोल्टेज (केवी) Voltage Level (kV)	दिनांक DATE	टिप्पणी Remarks	
NIL							
एच.वी.डी.सी/ए.सी फिल्टर बैंक/फैक्ट्स डिवाइस संबद्ध प्रणाली / HVDC /AC Filter bank / FACTS DEVICE associated System							
क्र. Sl. No.	एजेंसी/मालिक Agency/Owner	एलेमेंट का नाम Element Name	उप-केन्द्र SUB-STATION	वोल्टेज (केवी) Voltage Level (kV)	दिनांक DATE	टिप्पणी Remarks	
NIL							
बे / BAYS							
क्र. Sl. No.	एजेंसी/मालिक Agency/Owner	एलेमेंट का नाम Element Name	उप-केन्द्र SUB-STATION	वोल्टेज (केवी) Voltage Level (kV)	दिनांक DATE	टिप्पणी Remarks	
1	JSWEUL	400KV TIE BAY OF (OPGC-1 AND GT-2) AT JSWEUL	JSWEUL	400	19-01-2025		
2	BSPTCL	400KV MAIN BAY OF PATNA-1 AT BAKHTIYARPUR (BH)	BAKHTIYARPUR (BH)	400	18-01-2025		

3	WBPDC	400KV MAIN BAY OF 130 MVA ST-5 AT SAGARDIGHI	SAGARDIGHI	400	15-01-2025	
4	BSPTCL	220KV MAIN BAY OF GAYA(PG) -4 AT BODHGAYA	BODHGAYA	220	06-01-2025	
5	Power Department, Govt. of Sikkim	220KV MAIN BAY OF TASHIDING -1 AT LEGSHIP	LEGSHIP	220	01-01-2025	
6	BSPTCL	400KV MAIN BUS - 1 AT BAKHTIYARPUR (BH)	BAKHTIYARPUR (BH)	400	18-01-2025	
7	BSPTCL	400KV MAIN BAY OF BARH -1 AT BAKHTIYARPUR (BH)	BAKHTIYARPUR (BH)	400	31-01-2025	
8	BSPTCL	400KV MAIN BAY OF PATNA -2 AT BAKHTIYARPUR (BH)	BAKHTIYARPUR (BH)	400	31-01-2025	
9	BSPTCL	400KV MAIN BAY OF BARH -2 AT BAKHTIYARPUR (BH)	BAKHTIYARPUR (BH)	400	31-01-2025	
10	BSPTCL	400KV MAIN BAY OF PATNA -1 AT BAKHTIYARPUR (BH)	BAKHTIYARPUR (BH)	400	31-01-2025	

Members may note.

Deliberation in the meeting

Members noted.

4.4. UFR operation during the month of January 2025

Frequency profile for the month as follows:

MONTH	MAX	MIN	% LESS IEGC BAND	% WITHIN IEGC BAND	% MORE IEGC BAND
	(DATE/TIME)	(DATE/TIME)			
January, 2025	50.31 Hz on 24-01- 2025 at 14:01 hrs	49.60 Hz on 05-01- 2025 at 09:16 hrs	5.23	76.05	18.72

Hence, no report of operation of UFR has been received from any of the constituents.

Members may note.

Deliberation in the meeting

Members noted.

Annex A

Participants in 224th OCC Meeting

Venue: ERPC Conference Hall, Kolkata

Time: 10:30 Hrs.

Date: 25.02.2025 (Tuesday)

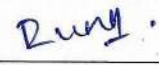
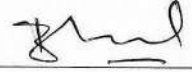
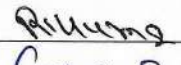
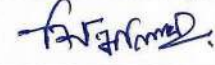
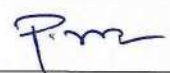
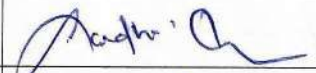
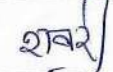
Sl.	Name	Designation	Organisation	Contact No.	E-mail Id	Signature
1	N S Mondal	Member Secretary	ERPC	9958389967	mserpc-power@nic.in	
2	R Sutradhar	Executive Director	ERLDC	9436302714	rajibsutradhar@grid-india.in	
3	Bilash Achari	DGM	ERLDC	7003472016	bilash.achari@grid-india.in	Bilash Achari
4	Manas Das	DGM	ERLDC, GRID-INDIA	9007070925	manasdas@grid-india.in	Manas Das
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6	S. K. Sharma	SGM	SLDGDV DVC LSC	9434539423	Sanjay.Sharma@dvc.gov.in	S K Sharma
7	Rajeev Pooasad	EEE(OS)	TVNL, Ranchi	8709054646	rajeev.pooasad@tvnl.in	Rajeev Pooasad
8	Krishna K. Bhattarai	CB (CHP)	DGPC Bhatra	+975 1762620	k.bhattarai266@drugreen.bt	Krishna K. Bhattarai
9	Pama Dorji	Unit Head (OP&M)	DGPC (MHP)	+975 1761975	p.dorji p.dorji 2735@drugreen.bt	Pama Dorji
10	P. V. RAUT.	AGM (OPN)	MPL	9223501513	rautpv@tatapower.com	P V Raut
11	Sumeet Narang	SGM(OS)	NTPC	8005493953	Sumeetnarang@ntpc.co.in	Sumeet Narang
12	Abhishek Murari	Deputy Mgr	NTPC, Baux	9470894493	ABHISHEK-MURARI@NTPC.CO.IN	Abhishek Murari
13	Phuntscho Choden	OS/THP	DGPC	+975 17905385	p.choden3586@drugreen.bt	Phuntscho Choden
14	Kezang Tshomo	Gr. SO (BHP)	DGPC (BHP)	+975 17918978	k.tshomo2458@drugreen.bt	Kezang Tshomo
15	Agniva Chatterjee	AD-I	ERPC	8100307502	agniva.cca@gov.in	Agniva Chatterjee

Participants in 224th OCC Meeting

Venue: ERPC Conference Hall, Kolkata

Time: 10:30 Hrs.

Date: 25.02.2025 (Tuesday)

Sl.	Name	Designation	Organisation	Contact No.	E-mail Id	Signature
16	Alok Pratap Sarm	Chief Manager	Grid India	9007285390	apsingh@grid-india.in	
17	Kumar Satyam	ACE	ERP	7355220042	Satyam.24365@gov.in	
18	P P Jena	EE	ERPC	9776198991	ppjena.erpcc@gov.in	
19	Bharendra Sahu	Manager	APNRL	9771430742	bharendra.sahu@adunikpower.co.in	
20	Ravi Kumar	Asst. Manager	APNRL	9971948707	ravikumarsarmad02@gmail.com	
21	CHANDAN MALLICK	Chief Manager	ERLDC, GRID-INDIA	7278756275	chandan.mallik@gridindia.in	
22	Santosh Kumar Panda	Manager	DVC, LDC	6370124294	santosh.panda@dvc.gov.in	
23	Abhishek Das	Assistant Consultant	TCS	9633835015	das.abhishek2@tcs.com	
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25	Rakesh Kr. Pradhan	Ch. Mgr.	GRID-INDIA	9831337570	rakesh.pradhan@grid-india.in	
26	Lalitha Kumar	Manager	GRID-INDIA, ERLDC	9831329428	lalitha@grid-india.in	
27	SAIBAL GHOSH	Manager	GRID-INDIA	8584072079	sailal@grid-india.in	
28	Saurav K. Sahay	DGM(MO)	GRID-INDIA	9432013173	Saurav.Sahay@grid-india.in	
29	Shabari Pramanick	CM	GRID-INDIA	9007058764	shabari.pramanick@grid-india.in	
30	MANJUNATHA M	EE	ERPC	9972223842	manjunatha.cea@gov.in	

Participants in 224th OCC Meeting

Venue: ERPC Conference Hall, Kolkata

Time: 10:30 Hrs.

Date: 25.02.2025 (Tuesday)

Sl.	Name	Designation	Organisation	Contact No.	E-mail Id	Signature
31	Sanatan Sarvesh	Deputy Director	CEA	9595219638	s.sarvesh@gov.in	S. Sarvesh
32	Gagan Kumar	EE CLDC	BSPTCL	7992496100	gagankumishra@gmail.com	Gagan Kumar
33	P. K. DE	ERPC SE	ERPC	9831620142	secommml.erpc@gov.in	P. K. DE
34	Pratham Kumar	EE	ERPC	8076873558	prathamkumar.ce@nic.in	Pratham Kumar
35	Sunil Kr Lutta	EE Sec	BSPTCL	7763817717	sunil2008itd@gmail.com	Sunil Kr Lutta
36	S. K. Maharana	G.M (Elec)	GRIDCO	9438591110	ele.skmaharana@gridco.co.in	S. K. Maharana
37	R. K. Biswal	DGM (RTM)	Power Grid	9078883641	sanjankumar@powergrid.in	R. K. Biswal
38	A. K. Ghosh	GM (OS)	WBPDCL	8336904026	akghosh@wbpdcl.co.in	A. K. Ghosh
39	Manoj Poddar	AGM (OS)	WBPDCL	8336904077	mpoddar@wbpdcl.co.in	Manoj Poddar
40	D. Chaki	CE, CPD	WBSETCL	9634910019	cpd.wbsetcl@gmail.com	D. Chaki
41	D Mukherjee	Sr. Mgr. (PS)	WBPDCL	9830052830	d.mukherjee@wbpdcl.co.in	D Mukherjee
42	DEBARSHI DE	SM (SO)	CEEC	9230521123	debarshi.de@ceec.in	Debarshi De
43	Raju Kachhap	Sr. Manager	SLDC, Ranchi	7783087568	rajmaifone82@gmail.com	Raju Kachhap
44						
45						

List of Participantsjoined online					
Name	First Join	Last Leave	In-Meeting Duration	Email	Role
ERPC Kolkata	2/25/25, 10:20:51 AM	2/25/25, 5:38:25 PM	3h 57m 31s	ERPC@KolkataMST.onmicrosoft.com	Organizer
Abhilash Gour (External)	2/25/25, 10:25:57 AM	2/25/25, 1:56:53 PM	3h 30m 55s	abhilash.gour@dansenergy1.onmicrosoft.com	Presenter
SLDC ODISHA (Unverified)	2/25/25, 10:25:59 AM	2/25/25, 11:23:13 AM	57m 13s		Presenter
Ch Mohan Rao, Powergrid, Odisha (Unverified)	2/25/25, 10:27:47 AM	2/25/25, 12:00:43 PM	1h 32m 56s		Presenter
Madhumita Soren (External)	2/25/25, 10:30:33 AM	2/25/25, 1:08:01 PM	2h 37m 28s	madhumita.soren@opgc.co.in	Attendee
JLHEP (Unverified)	2/25/25, 10:30:33 AM	2/25/25, 1:53:41 PM	3h 23m 8s		Presenter
PRASANNA SAHOO (Unverified)	2/25/25, 10:30:34 AM	2/25/25, 1:07:59 PM	2h 37m 24s		Presenter
Debanjan Sarkar {देबनजान सरकार} (External)	2/25/25, 10:30:34 AM	2/25/25, 12:00:19 PM	1h 29m 44s	debanjan@powergrid.in	Presenter
SGM (ELE) , Corporate Planning (OPTCL) (Unverified)	2/25/25, 10:31:48 AM	2/25/25, 12:39:54 PM	2h 8m 5s		Presenter
Mamidi Prasad {मामिडी प्रसाद} (External)	2/25/25, 10:44:23 AM	2/25/25, 12:09:58 PM	1h 25m 34s	mamidi.prasad@powergrid.in	Presenter
Ch Eswara Rao (Unverified)	2/25/25, 11:07:38 AM	2/25/25, 12:00:00 PM	52m 22s		Presenter
SLDC ODISHA (Unverified)	2/25/25, 11:23:33 AM	2/25/25, 5:38:25 PM	6h 14m 52s		Presenter
Gautam Ranjan (External)	2/25/25, 12:02:41 PM	2/25/25, 3:00:02 PM	2h 57m 20s	GAUTAMRANJAN@NTPC.CO.IN	Presenter

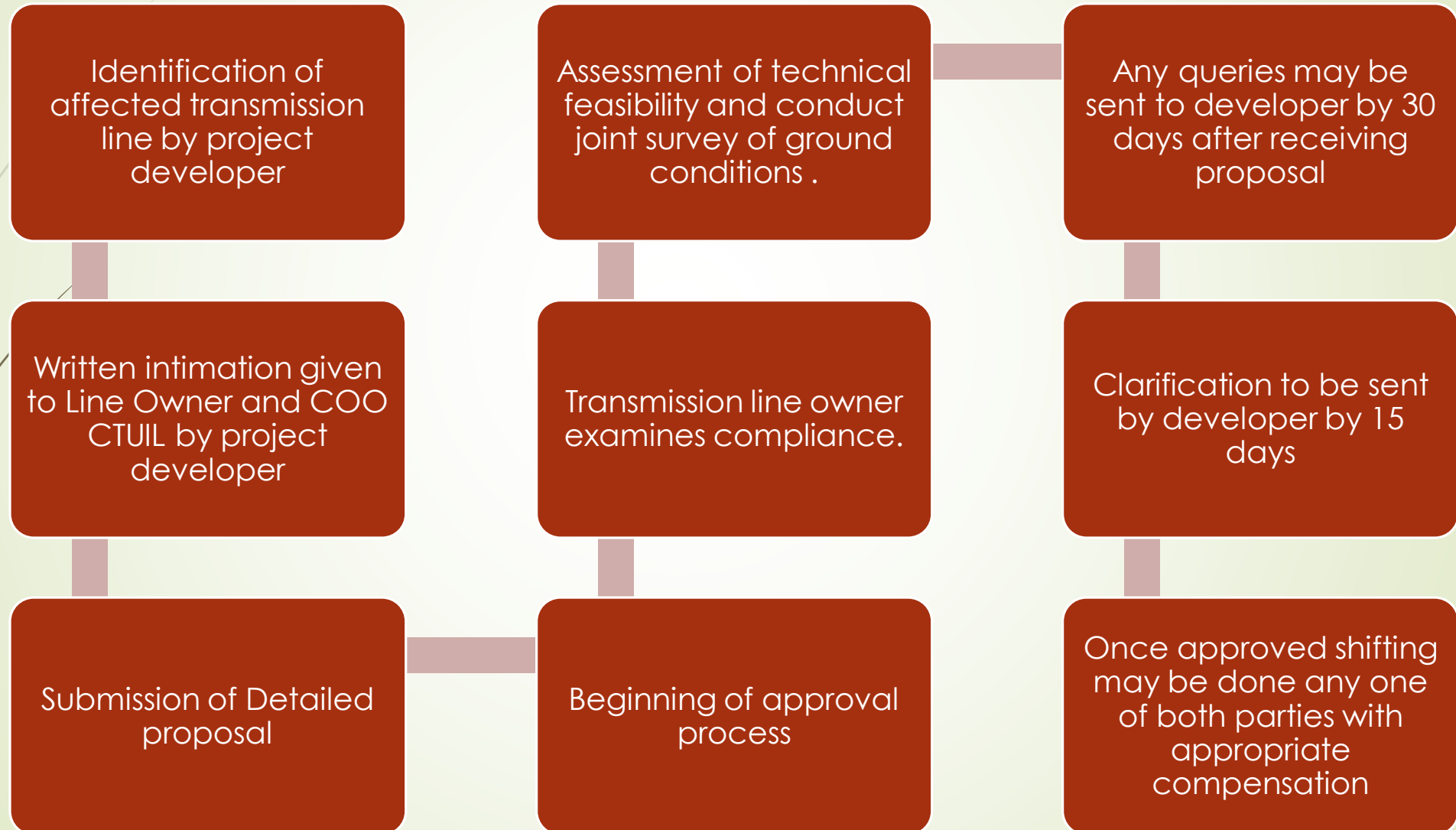


SOP on Shifting of Transmission Lines for other infrastructure projects.

Introduction

- Several infrastructure projects are being undertaken by NHAI, BRO, Indian Railways etc which include – roads, highways, airports, mines etc.
- These projects often come across existing/under construction transmission lines in their route alignment, leading to the need for shifting of such lines for construction of these projects.
- Projects monitored on the portals of Department for Promotion of Industry and Internal Trade (DPIIT) (Project Monitoring Group) or Ministry of Statistics and Program Implementation (MoSPI) shall be considered as projects of National Importance.
- Due to inadequate coordination between Project Developers and owners of transmission lines - Transmission Service Providers/ Transmission licensees - frequently compromises the safety of transmission lines during construction and leads to significant delays in project completion.
- To address these challenges, ensure compliance with existing regulatory provisions issued by the Central Electricity Authority (CEA) and other authorities a Standard Operating Procedure (SOP) for shifting of Transmission lines by various Project Developers had been notified by CEA dated 10.03.2023

Application and Approval for shifting Transmission Line



Shifting/Alteration work done by Project Developer

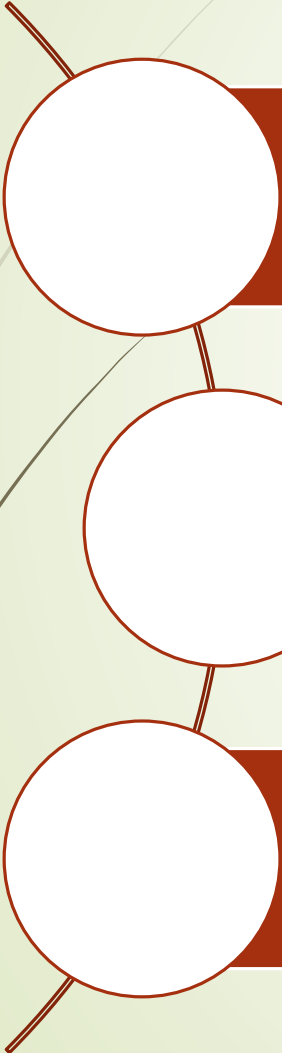
All design documents to be submitted to line owner by developer. Owner shall review and then provide final approval within 3 weeks, only then work will commence. An MoU is to be signed by both parties that outlines mutually agreed terms and conditions

Cost for the work to be borne by project developer. Supervision charges to be paid to owner prior to commencement of work

All permissions such as ROW compensations, Forest Clearance, Wildlife clearance to be taken by developer

Approval of shutdowns required during work to be taken from RPC or SLDC

Shifting/Alteration work done by Owner of Transmission Line



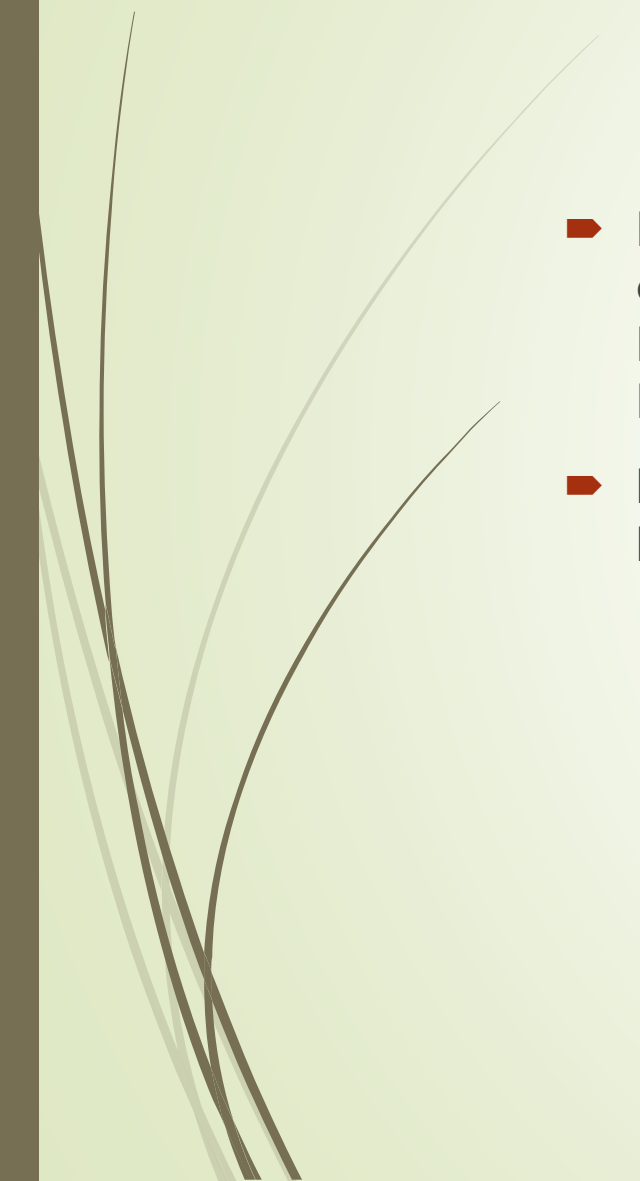
Owner shall calculate all costs (direct and indirect) and communicate the cost estimate to the project developer after approval by 30 days

Project developer to submit estimated cost to Owner. Final Decision for any dispute of cost estimate will be done taken by COO CTUIL. Parties may then execute MoU outlining mutually agreed terms and conditions.

All shifting work to be completed within **8 months** of the first request of the project developer.



Design and Standards

- Design, testing, construction and erection of transmission lines shall be in accordance to CEA (Technical Standards for Construction of Electrical Plants and Electric Lines) Regulations, CEA (Measures related to Safety and Electric Supply) Regulations.
 - In addition applicable guidelines of other authorities such as NHAI, BRO, Indian Railways may be followed.
- 



Availability of lines under shifting and alteration

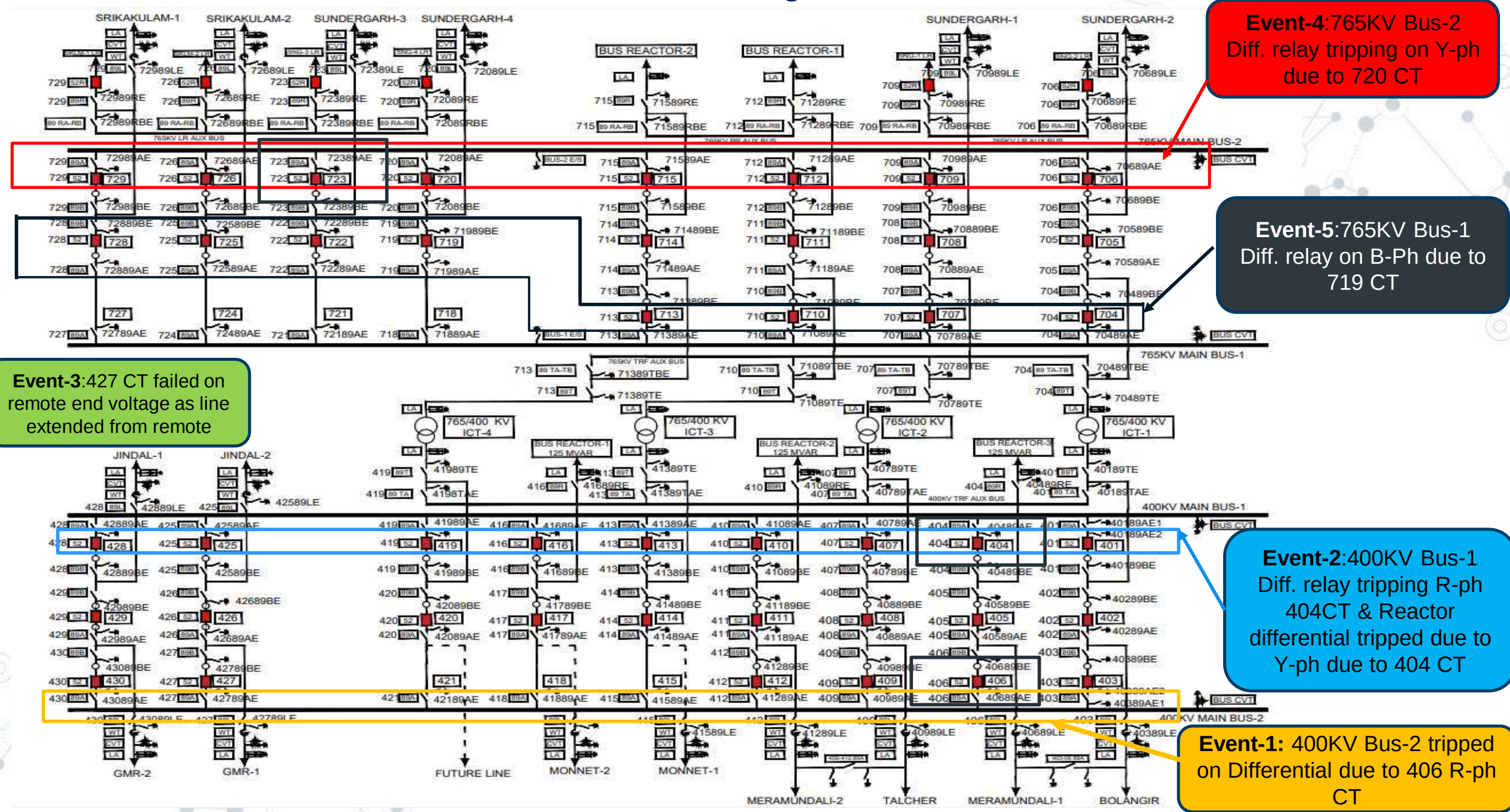
- ▶ RPC shall provide deemed availability certificate for the shutdown period of transmission licensee (RTM, TBCB) for shifting of ISTS lines.
- ▶ For cases where deemed availability certificate is not issued to transmission licensee for the shutdown period then charges toward loss of availability shall be borne by Project developer.

Report on trippings at Angul on 20.02.25

Preliminary report on Multiple element trippings at Angul on 20.02.25

STATION:	765/400KV Angul S/s
Tripping event Date & Time	20/02/2025 16:19:50 hrs
Details	765KV Bus-1 & 2 and 400KV Bus-1 &2 Tripped during Heavy Whirl Wind, Hail storm & Lighting.
Condition prior incident	▪ 765kV Angul-Sundergarh Line-4 was out for Voltage regulation & its corresponding LR was charged as Bus Reactor. ▪ 765 KV Bus Reactor-1 was out of service for Voltage regulation and Its Main Bay was out for Scheduled AMP Work. ▪ 400 KV Angul-Meramundali-1 was anti-theft charged. ▪ 400 KV Talcher & Meramundali-2 were in LILO bypass mode.
Elements tripped	16:19:50:018- 400 KV Bus-2 tripped on R-ph Differential 16:20:20:536- 400 KV Bus-1 tripped on R-ph Differential 16:20:23:302- 427 LBB Optd 16:20:27:751- 765 KV Bus-2 tripped on Y-ph Differential 16:20:31:186- 723 LBB Optd 16:20:55:210- 765 KV Bus-1 tripped on B-ph Differential

Angul SLD



Sequence of Events

Time	Trip element	Description	Remarks
16:19:50:019	400 KV Bus-2 R-ph	Fault in 406 Bay CT Fault current: 25.6kA	405 CB opened after 200 msec as Fault current persisting in CT after 406CB opened.
16:20:20:536	400 KV Bus-1 R-ph & 400Kv B/R-3	400 KV Bus-1 tripped on R-ph (Fault current:10.3kA) & Bus reactor -3 trpped on Y-ph Differential due to 404 Bay (Fault current:21.6kA)	
16:20:23:302	427 LBB Optd	After Tripping of Both Bus of 400 KV Bus Bar, 427 Bay CT was charged from remote end i.e. GMR Ckt-1. LBB optd in this bay.	B-ph of 427 Bay CT failed and LBB optd hence 426 CB opened
16:20:27:751	765 KV Bus-2 Y-ph	Bus diff in Y-ph CT of 720. Fault current:11.67kA	After 200 msec 719 CB opened after 200 msec as Fault current persisting in CT after 720CB opened. Line was already in open condition on Voltage regulation and its LR was in service as BR.
16:20:31:186	723 LBB Optd	723 CT Failed and LBB operated. Fault current:10.75kA.	As 723 Bay was tripped due to Main Bus-2 Fault but 723 CT was charged through Main Bus-1/ Line Side. after 200 msec its Tie CB 722 Got tripped thus the corresponding Angul-Sundergarh Line-3
16:20:55:210	765 KV Bus-1 B-ph	Bus differential in 719 B-ph CT. Fault current: 9.75kA	

Sequence of Restoration:

- i. 17:57 - 765 kV Main Bus #2(By charging 409 BAY)
- ii. 18:09 - BUS Reactor -2 taken into service.
- iii. 18:15 - 400 kV Main Bus #2.
- iv. 18:23 - 400 kV GMR #2
- v. 18:29 - 400 kV JITPL #1
- vi. 18:46 - ICT #4 charged from 765 kV side
- vii. 18:47 - ICT #4 charged from 400 kV side
- viii.18:51 - 400 kV Bus #1
- ix. 18:57 – 400 kV Jindal #2
- x. 19:00 – 400 kV Main Bay of Bolangir
- xi. 19:05 - ICT #1 charged from 400 kV side
- xii. 19:07 – 765 kV Main Bay of Sng #2
- xiii.19:24 – BR #1 charged
- xiv.19:35 – BR #2 charged
- xv. 19:46 – 400 kV Main Bay of ICT #2

Sequence of Restoration:



- xvi. 19:48 - 400 kV Main Bay of ICT #4
- xvii 19:55 – 765 kV Sklm #2
- xviii 20:25 – 400 kV Tie Bay of Jindal #2 & GMR #1
- xix 20:31 – 765 kV Sklm #1
- xx. 23:26 -1500 MVA ICT-3
- xxi 00:02-765kV Angul-Jharsuguda CKT-3
- xxii 00:02-765kV BUS-1 charged
- xxiii 00:08-Tie Bay of 765 kV Srikakulam-2
- xxiv 00:08- Tie Bay of 765 kV Srikakulam-1
- xxv 00:13- Main Bay of 1500 MVA ICT-4
- xxvi 00:14- Main Bay of 1500 MVA ICT-3
- xxvii 00:15- Main Bay of 1500 MVA ICT-1
- xxviii 00:15- Main Bay of 1500 MVA ICT-2

To be restored : 765 KV -Bay No-719,720 and 723 & 400kV- 404,405,406 and 427



720CT



427CT



406CT



External view of CTs



Maintenance by NTPC Plants

	Bus					ICT					Line (For Line Bay)				
Plant Name	Total Available	2021	2022	2023	2024	Total Available	2021	2022	2023	2024	Total Available	2021	2022	2023	2024
BARH	4				2	3	2	3	3	3	8	4	4	4	8
DARJIPALI	2				2	2				2	2				0
FSTPP	2				1	2		2	1	0	10	5	10	4	1
KHSTPP	4				0	2				0	12	8	1	1	3
NABINAGAR(BRBCL)	2				0	3				0	2				0
NABINAGAR(NPGC)	2	1	1		0	3		1	1	0	4				1
TSTPP	6				0	2	1			0	10	2	1	3	0

Annexure D.1

Updated Anticipated Peak Demand (in MW) of ER & its constituents for March 2025

1	BIHAR	Demand (MW)	Energy Requirement (MU)
	NET MAX DEMAND	5709	3530
	NET POWER AVAILABILITY- Own Sources	1253	408
	Central Sector+Bi-Lateral	4542	4410
	SURPLUS(+)/DEFICIT(-)	86	1287
2	JHARKHAND		
	NET MAXIMUM DEMAND	1915	1095
	NET POWER AVAILABILITY- Own Source	330	201
	Central Sector+Bi-Lateral+IPP	1325	764
	SURPLUS(+)/DEFICIT(-)	-260	-130
3	DVC		
	NET MAXIMUM DEMAND	3200	2215
	NET POWER AVAILABILITY- Own Source	5700	3512
	Central Sector+MPL	300	154
	Bi- lateral export by DVC	2400	1806
	SURPLUS(+)/DEFICIT(-) AFTER EXPORT	400	-356
4	ODISHA		
	NET MAXIMUM DEMAND (OWN)	5250	3534
	NET MAXIMUM DEMAND (In Case of CPP Drawal of 900 MW(peak) and average drawl of 700 MW)	6150	4055
	NET POWER AVAILABILITY- Own Source	3246	2188
	Central Sector	2050	1380
	SURPLUS(+)/DEFICIT(-) (OWN)	46	34
	SURPLUS(+)/DEFICIT(-) (I(In Case of CPP Drawal of 900 MW(peak) and average drawlm of 700 MW)	-854	-487
5	WEST BENGAL		
	WBSEDCL		
5.1	NET MAXIMUM DEMAND	6650	4519
	NET MAXIMUM DEMAND (Incl. Sikkim)	6655	4523
	NET POWER AVAILABILITY- Own Source (Incl. DPL)	5125	3317
	Central Sector+Bi-lateral+IPP&CPP+TLDP	2652	1289
	EXPORT (To SIKKIM)	5	4
	SURPLUS(+)/DEFICIT(-) AFTER EXPORT	905	82
5.2	CESC		
	NET MAXIMUM DEMAND	1950	966
	NET POWER AVAILABILITY- Own Source	700	537
	IMPORT FROM HEL	540	350
	TOTAL AVAILABILITY OF CESC	1950	887
	SURPLUS(+)/DEFICIT(-)	0	-79
	WEST BENGAL (WBSEDCL+CESC+IPCL)		
	(excluding DVC's supply to WBSEDCL's command area)		
	NET MAXIMUM DEMAND	8468	5485
	NET POWER AVAILABILITY- Own Source	5825	3854
	CS SHARE+BILATERAL+IPP/CPP+TLDP+HEL	3193	1639
	SURPLUS(+)/DEFICIT(-) BEFORE WBSEDCL'S EXPORT	551	7
	SURPLUS(+)/DEFICIT(-) AFTER WBSEDCL'S EXPORT	546	3
6	SIKKIM		
	NET MAXIMUM DEMAND	129	62
	NET POWER AVAILABILITY- Own Source	44	48
	Central Sector	308	184
	SURPLUS(+)/DEFICIT(-)	223	170
	EASTERN REGION		
	NET MAXIMUM DEMAND	24803	15921
	NET MAXIMUM DEMAND ((In Case of CPP Drawal of 800 MW(peak) and average drawl of 700 MW)	25703	16442
	BILATERAL EXPORT BY DVC (Incl. Bangladesh)	2415	1806
	EXPORT BY WBSEDCL TO SIKKIM	5	4
	EXPORT TO B'DESH & NEPAL OTHER THAN DVC	642	431
	NET TOTAL POWER AVAILABILITY OF ER	26478	17668
	(INCLUDING CS ALLOCATION +BILATERAL+IPP/CPP+HEL)		
	SURPLUS(+)/DEFICIT(-)	-1386	-495
	SURPLUS(+)/DEFICIT(-) (In Case of CPP Drawal for Odisha)	-2286	-1016